

TTK4240 Industriell elektroteknikk

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Marta Molinas

Institutt for teknisk kybernetikk



DC Motors

Based on Allan R. Hambley,
Electrical Engineering Principles and Application, 4th ed.,
Chapter 16

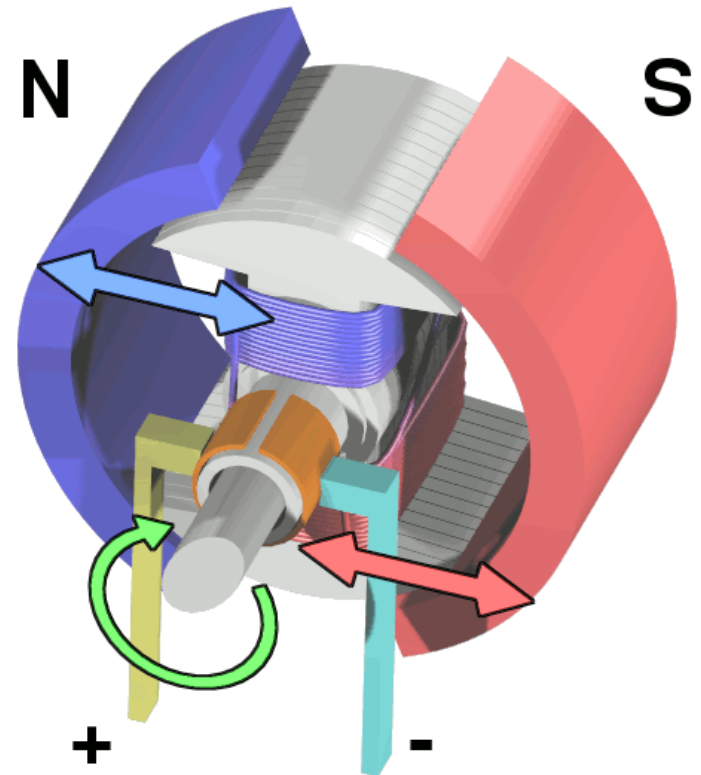
Lecture - Week 08 – DC Motor

Working principle of DC Motor

- Idealized linear motor
- Rotating DC motor
- Induced EMF and commutation
- Equivalent Circuit of DC motor
- Types of DC motors
 - Shunt connected motors
 - Separately excited motors
 - Permanent magnet motors
 - Series connected motors
- Torque-speed characteristics
- Brushless DC Motor

• Practical aspects

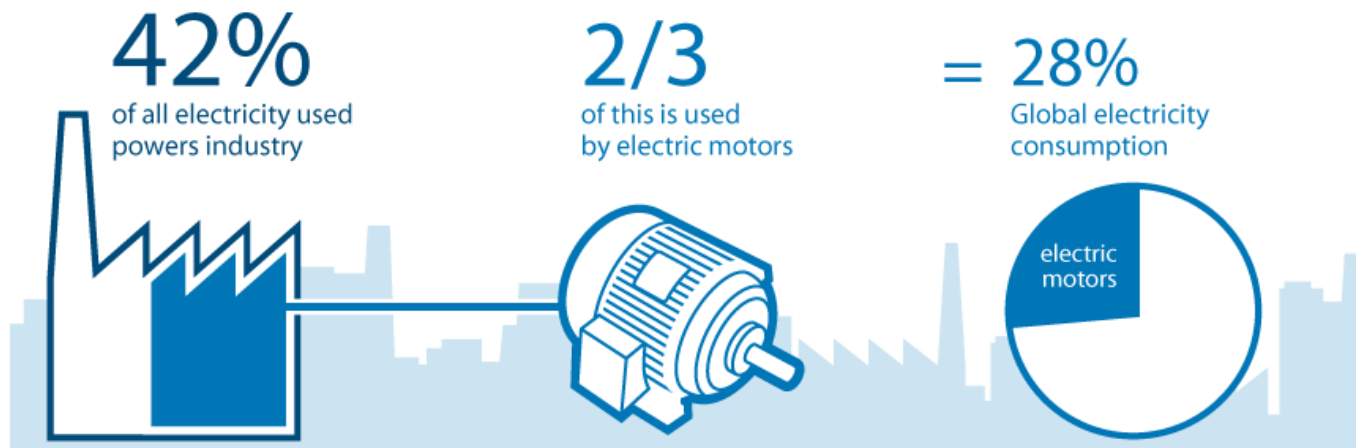
- Nameplate
- Datasheet



Motors convert electrical energy into mechanical energy

All actuators use some kind of motor

Much of this electricity is used to power industrial electric motors



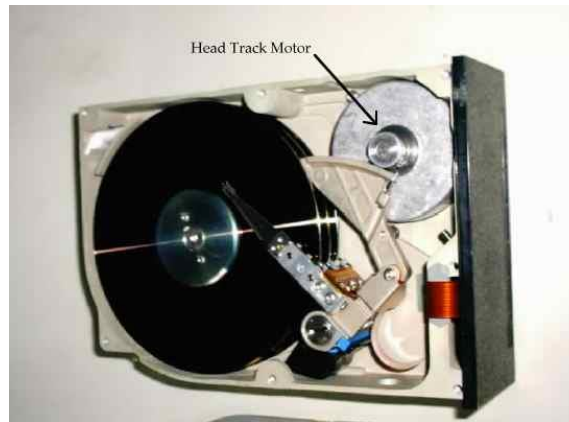
DC Motor: where it is used

Low
power

- **Everyday life:** computer disks, refrigerators, garage-door openers, washing machines, food mixers, vacuum cleaners, DVD players, ventilation fans, automotive power windows, windshield wipers, elevators, robotics, etc

High
power

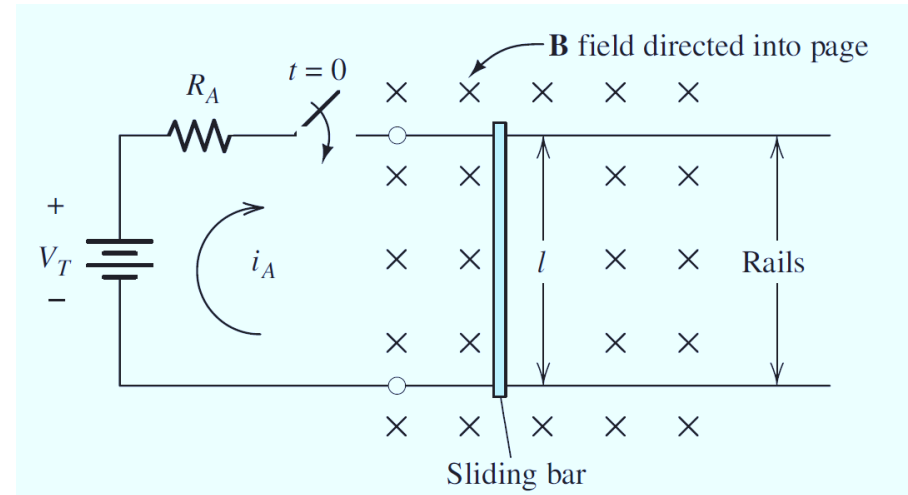
- **Industrial applications:** bulk material handling, machining operations, pumps, rock crushers, fans, compressors, and hoists



Ideal and Linear DC Motor

Idealized linear motor

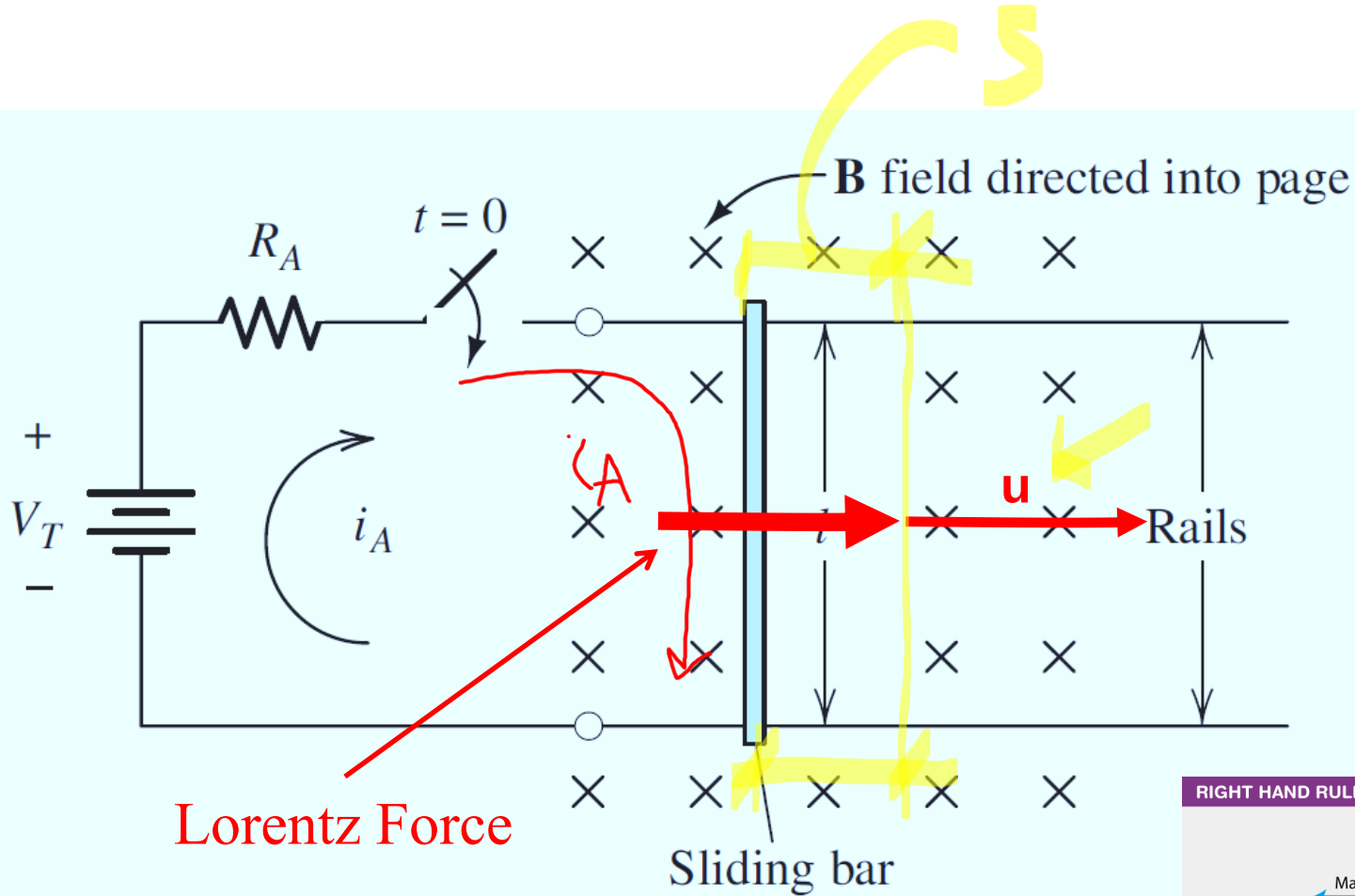
- An ideal and linear motor is first introduced to better understand the following concepts and principles of a rotating dc motor:
 - What is the **back electromotive force** (EMF)
 - What is the torque-speed characteristic
 - How to change the operating mode from motor to generator



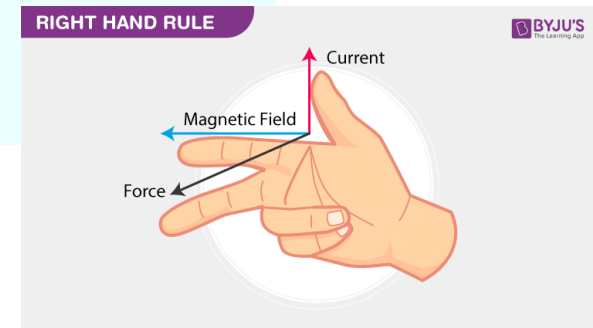
Idealized linear motor:

- The rails and the bar have zero resistance.
- A magnetic field is directed into the page, perpendicular to the plane of the rails and the bar.
- No frictions on the rails.

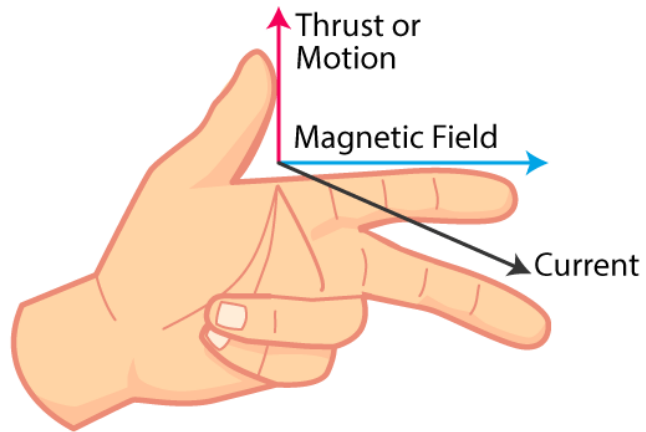
Idealized linear motor: working principle



Scenario: $u=0$ and switch closes at $t=0$

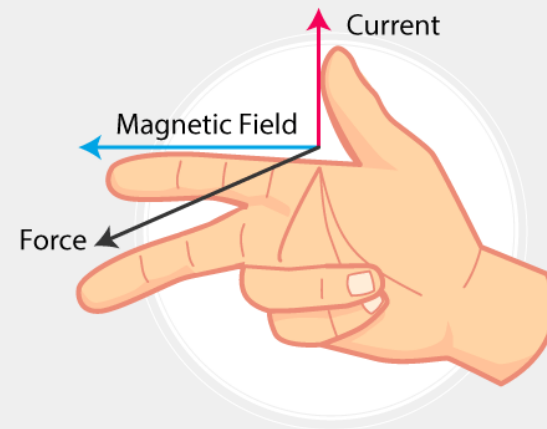


FLEMING'S LEFT HAND RULE



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RIGHT HAND RULE



Idealized linear motor

Current:

$$i_A(0+) = V_T / R_A$$

Force on the conductor:

$$f = i_A l B$$

⇒ Rightward movement

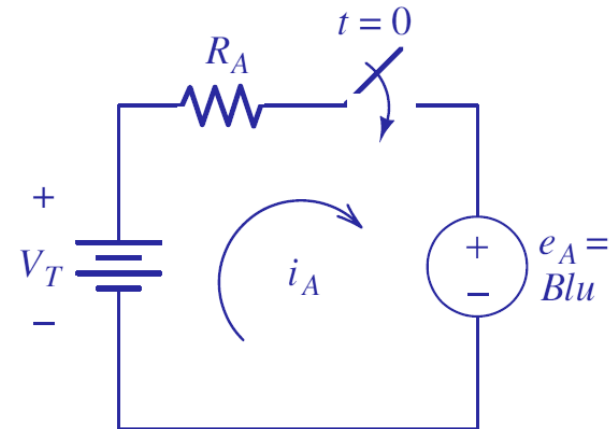
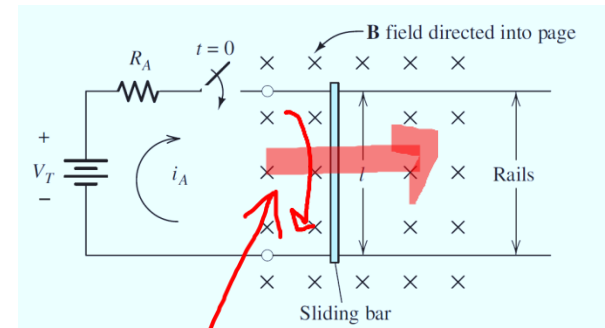
Induced voltage:

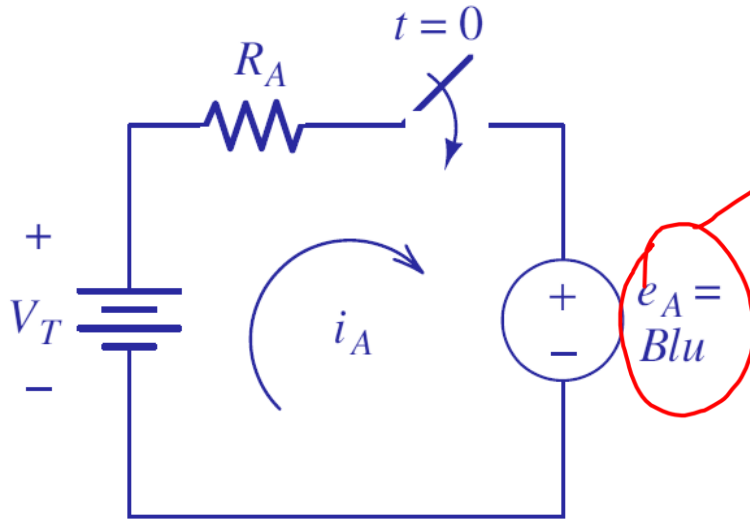
$$e_A = \frac{d\phi}{dt}$$

$$e_A = Blu$$

⇒ Reduced current

$$i_A = \frac{V_T - e_A}{R_A}$$





$$i_A = \frac{V_T - e_A}{R_A}$$

$$e_A = V_T$$

No friction
No resistance

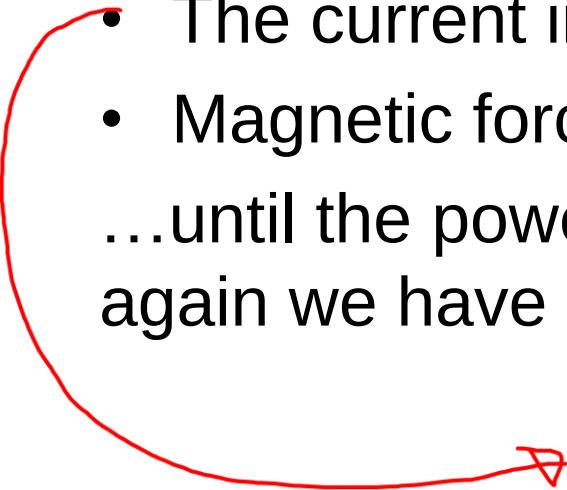
As velocity builds up, energy is absorbed by the induced voltage e_A , and this energy shows up as kinetic energy of the bar. Eventually, the bar speed becomes high enough that $e_A = V_T$. Then, the current and the force become zero, and the bar coasts at constant velocity.

Idealized linear machine: motor mode

If we attach a load to the rod:

- Power from the load reduces the velocity $\mathbf{u}$
- The induced voltage decreases $e_A = Blu$
- The current in the motor increases
- Magnetic force on the rod increases $f = i_A l B$

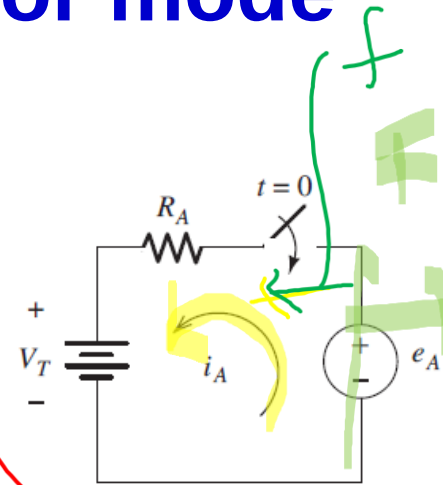
...until the power equals the load power. Then again we have stationary conditions, $\mathbf{u}$ is constant


$$i_A = \frac{V_T - e_A}{R_A}$$

Idealized linear machine: generator mode

If we actively pull the rod with constant force towards the right:

- The power makes the velocity u to increase
- The induced voltage exceeds the battery voltage
- Current circulates counter clockwise and charges the battery!
- The magnetic force on the rod increases to the left ...until it outweighs the power we exert on the rod.
Then again we have stationary condition and u is constant



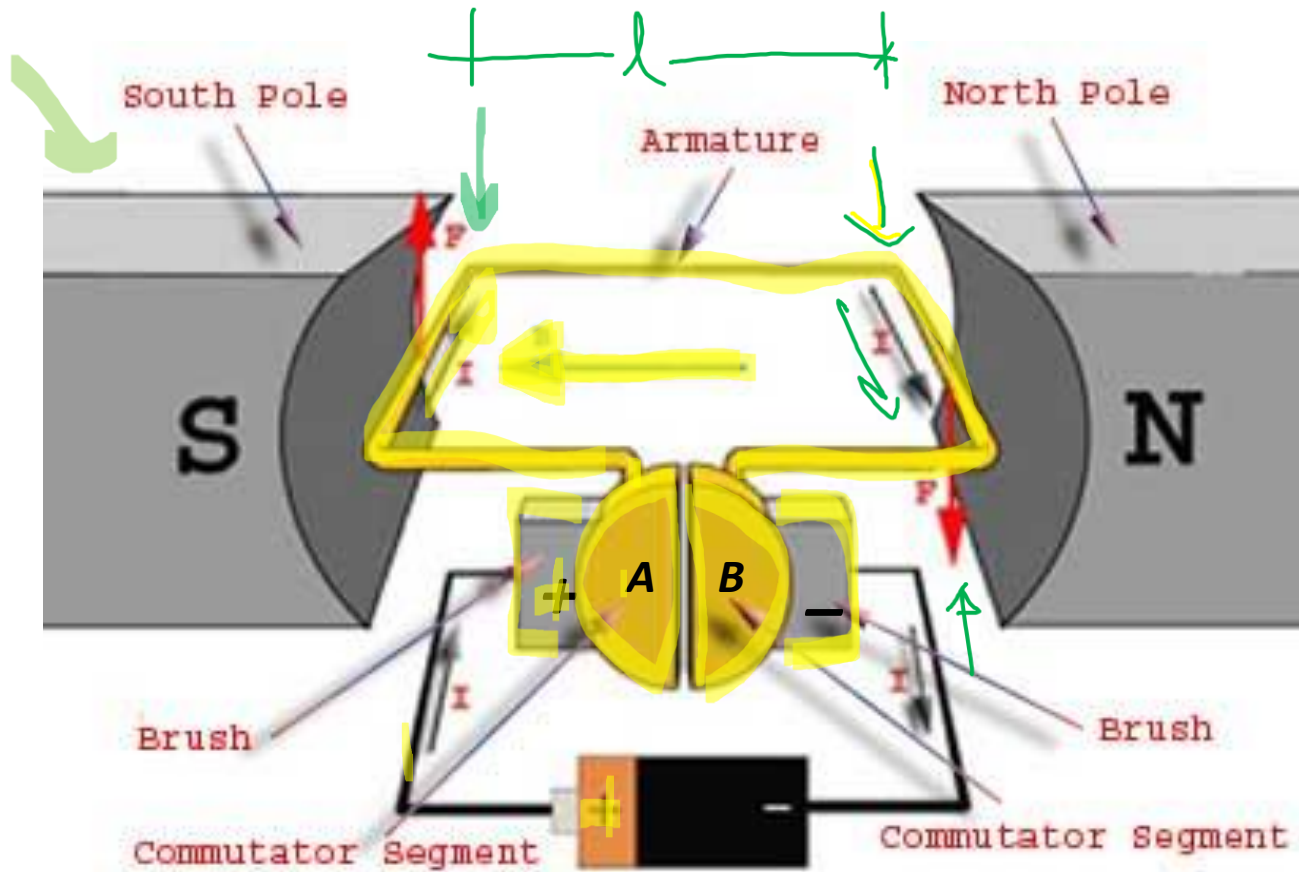
$$i_A = \frac{V_T - e_A}{R_A}$$

$$(-i_A)$$

$$e_A > V_T$$

Rotating DC Motors

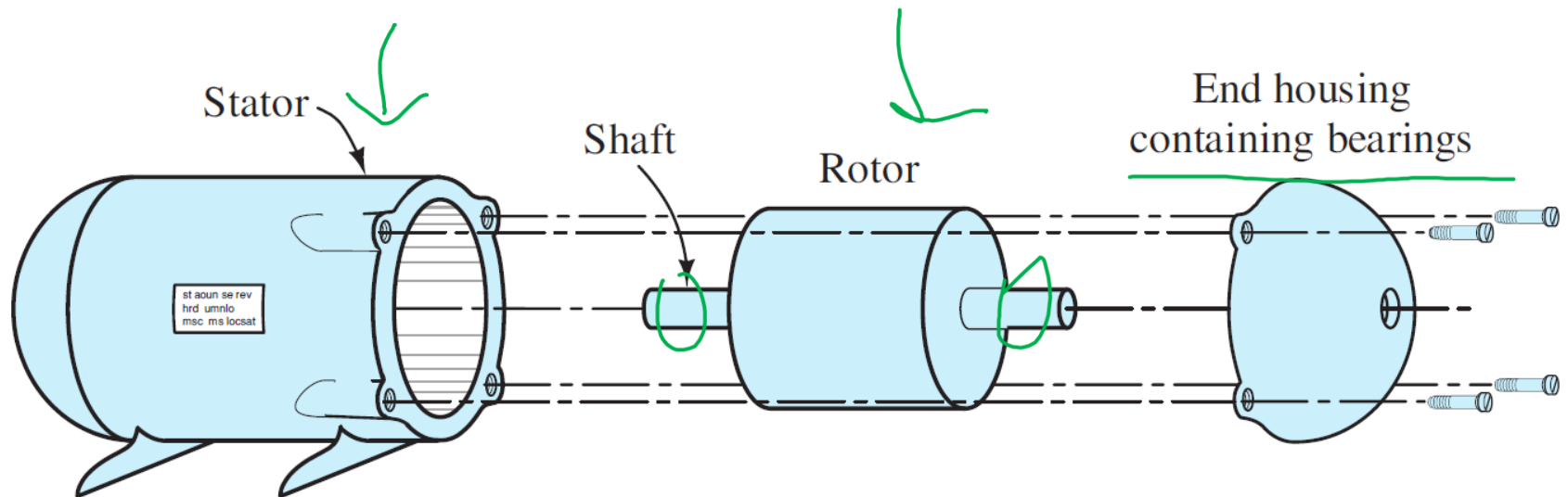
Schematic of a rotating DC motor



$$\underline{\underline{T = F \times l}}$$

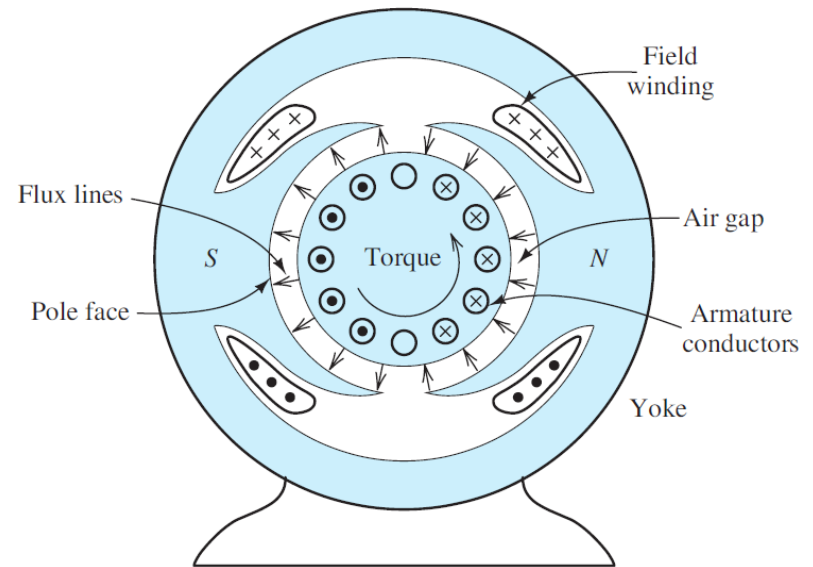
DC Motor Construction

DC machine basic construction



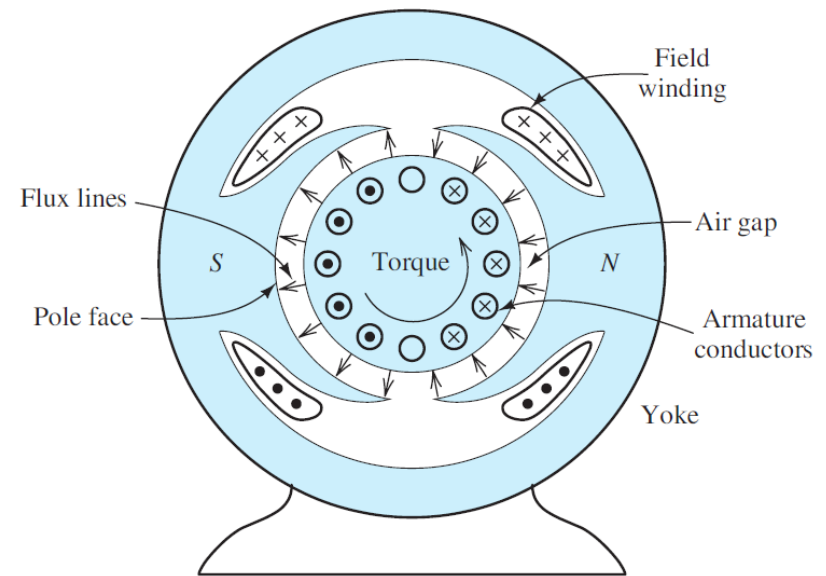
Field Windings

- Establishes the main magnetic field in the motor (electromagnet)
- The current is the same independent of motor load
- May be in stator or rotor

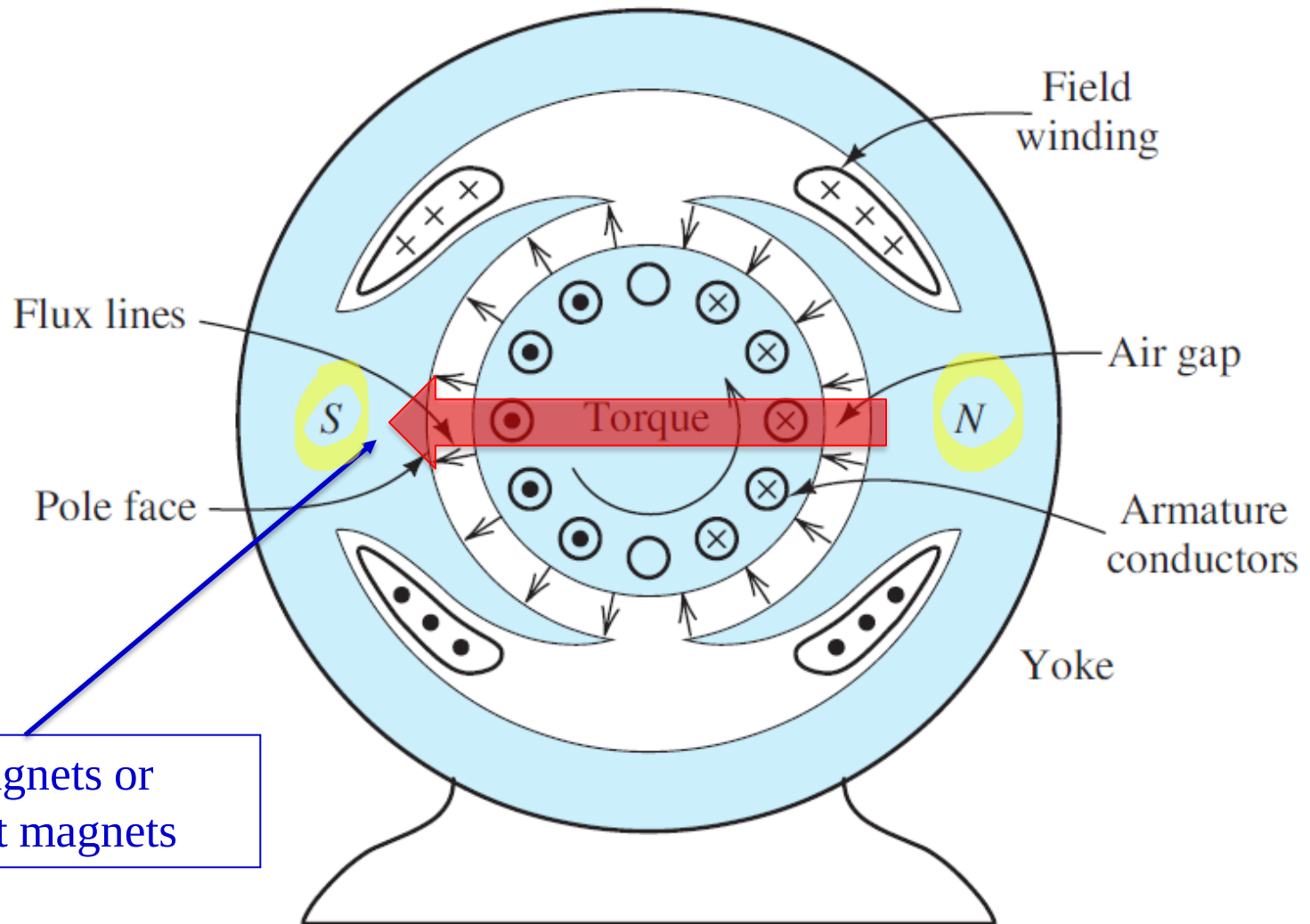


Armature Winding

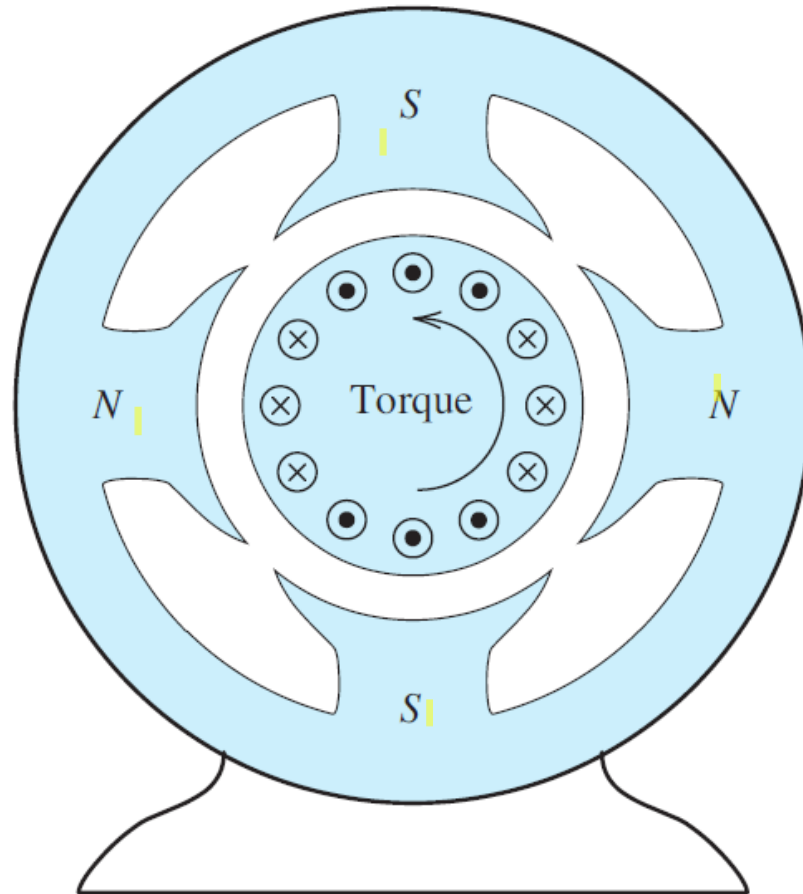
- Windings on the rotor that carries currents.
- It is the armature winding that carries the load
 - Low current at low motor load,
 - Large current with large motor load



Currents and Field– 2 pole machine

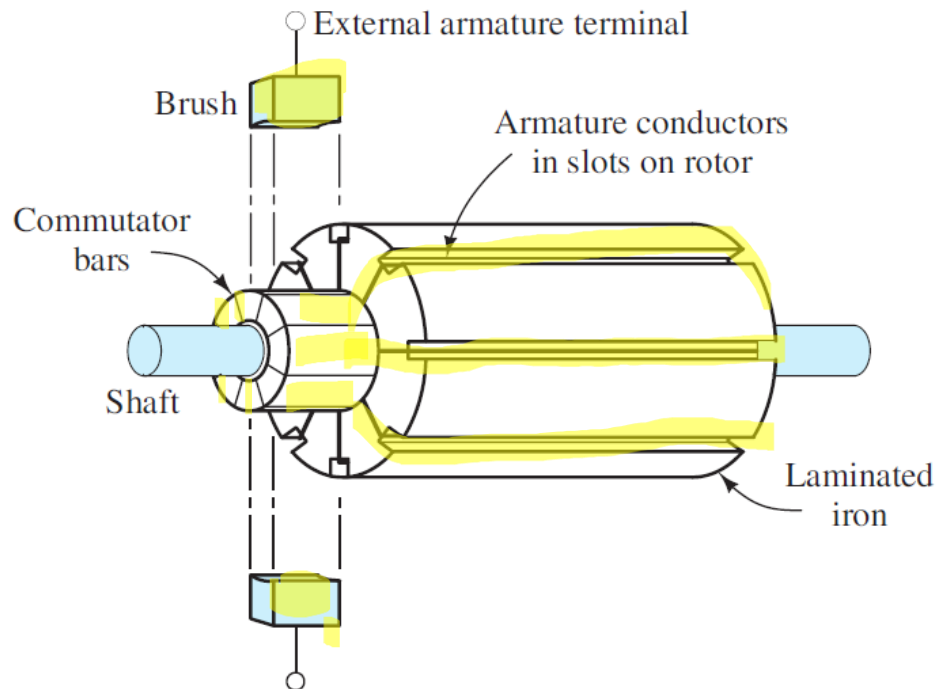


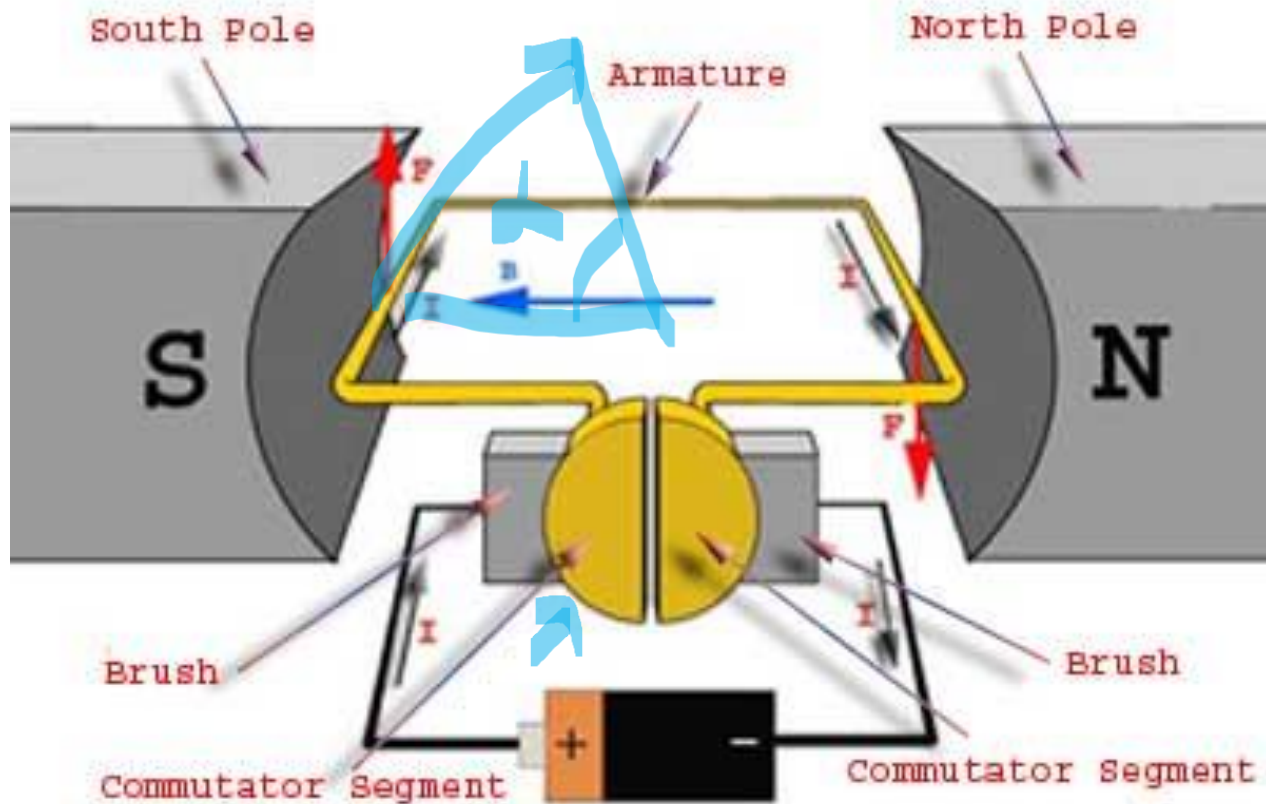
Four-pole machine



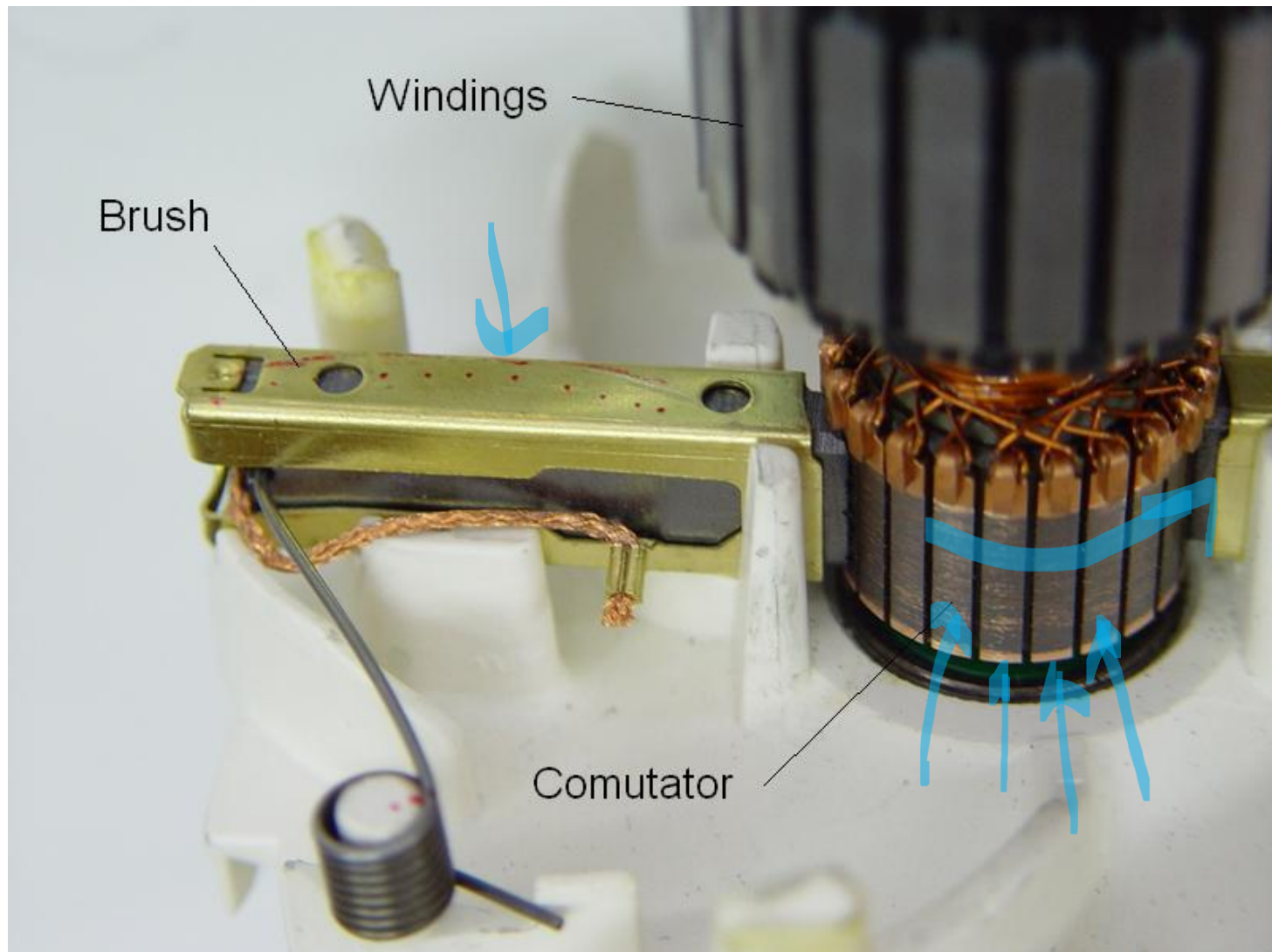
Commutator («current shifter»)

Mechanical "switch" that provides for turning the flow direction when armature winding moves from one pole to the next.

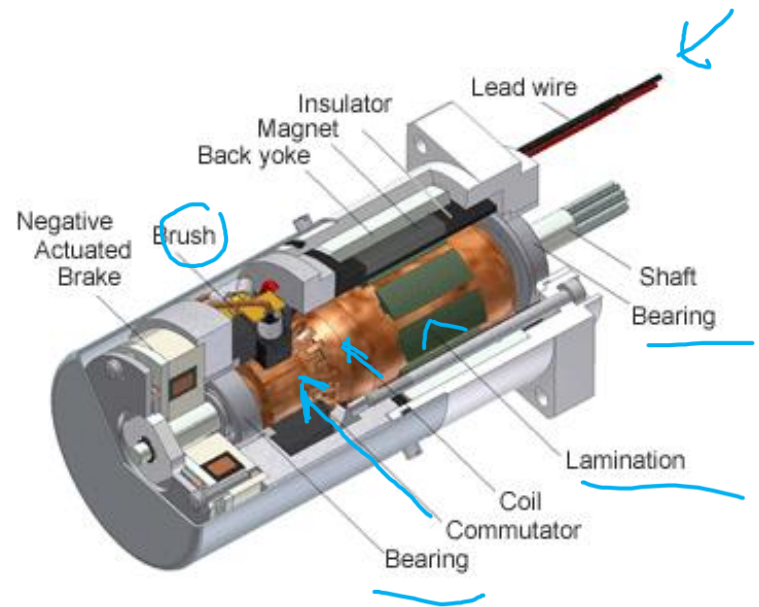
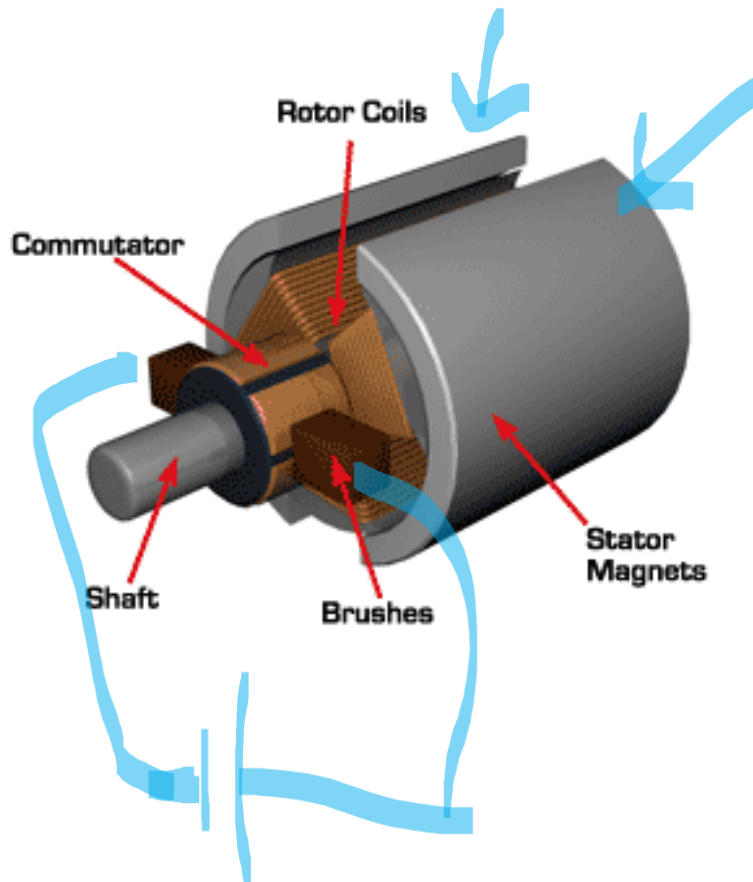




Commutator



Rotor construction



Working principle: Torque (Video)



$v = \text{constant}$

$$e_A = \frac{d\phi}{dt}$$

$$\phi = BA = B \cdot x l$$

$v = \text{const}$

$$e_A = \frac{d\phi}{dt} = \frac{d(B x l)}{dt} = B l \left(\frac{dx}{dt} \right)$$

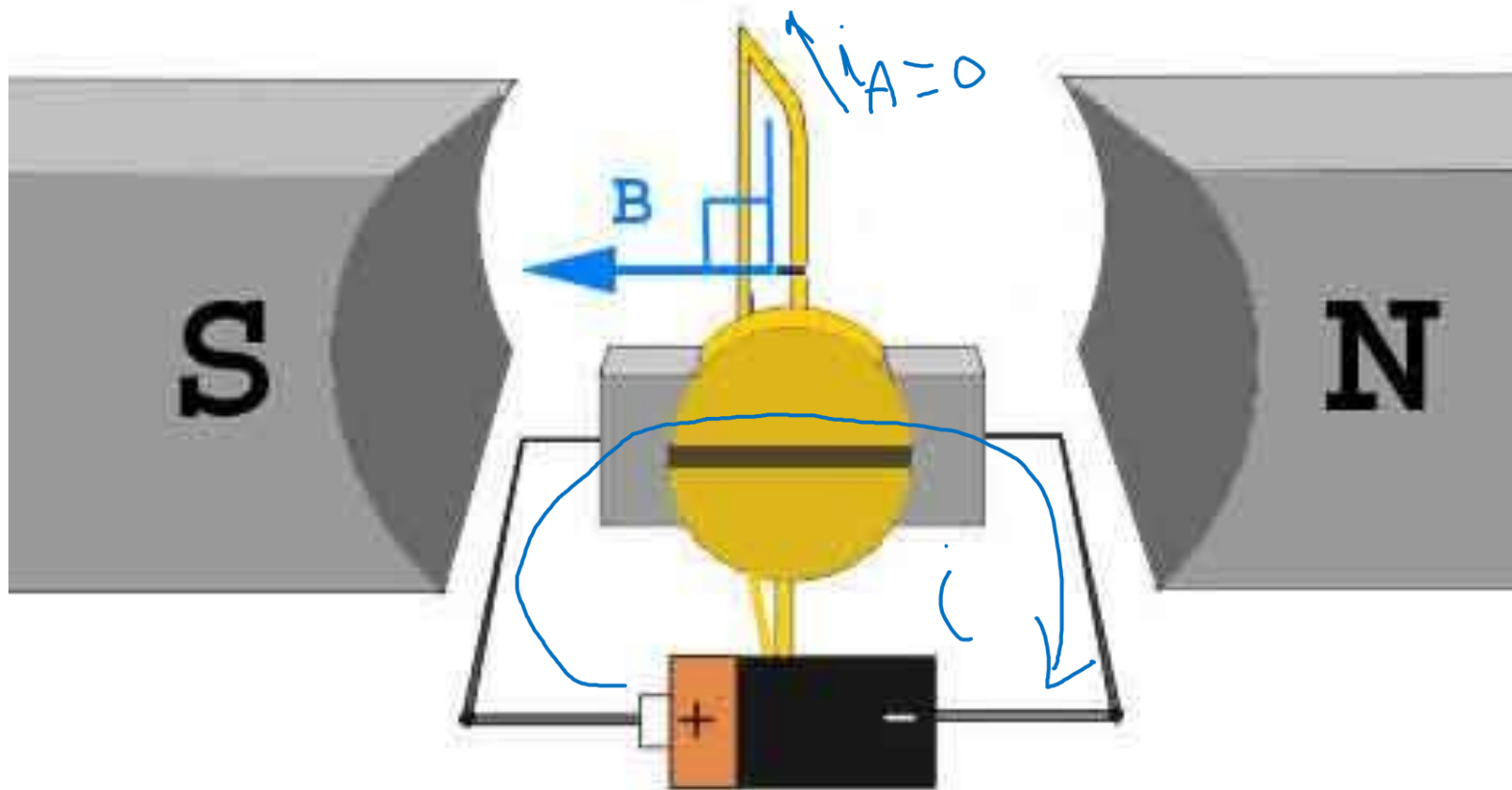
$$e_A = \overset{\downarrow \downarrow \downarrow}{B l v} = \text{constant}$$

$$i_A = \frac{V_T - e_A}{R_{A1}} =$$

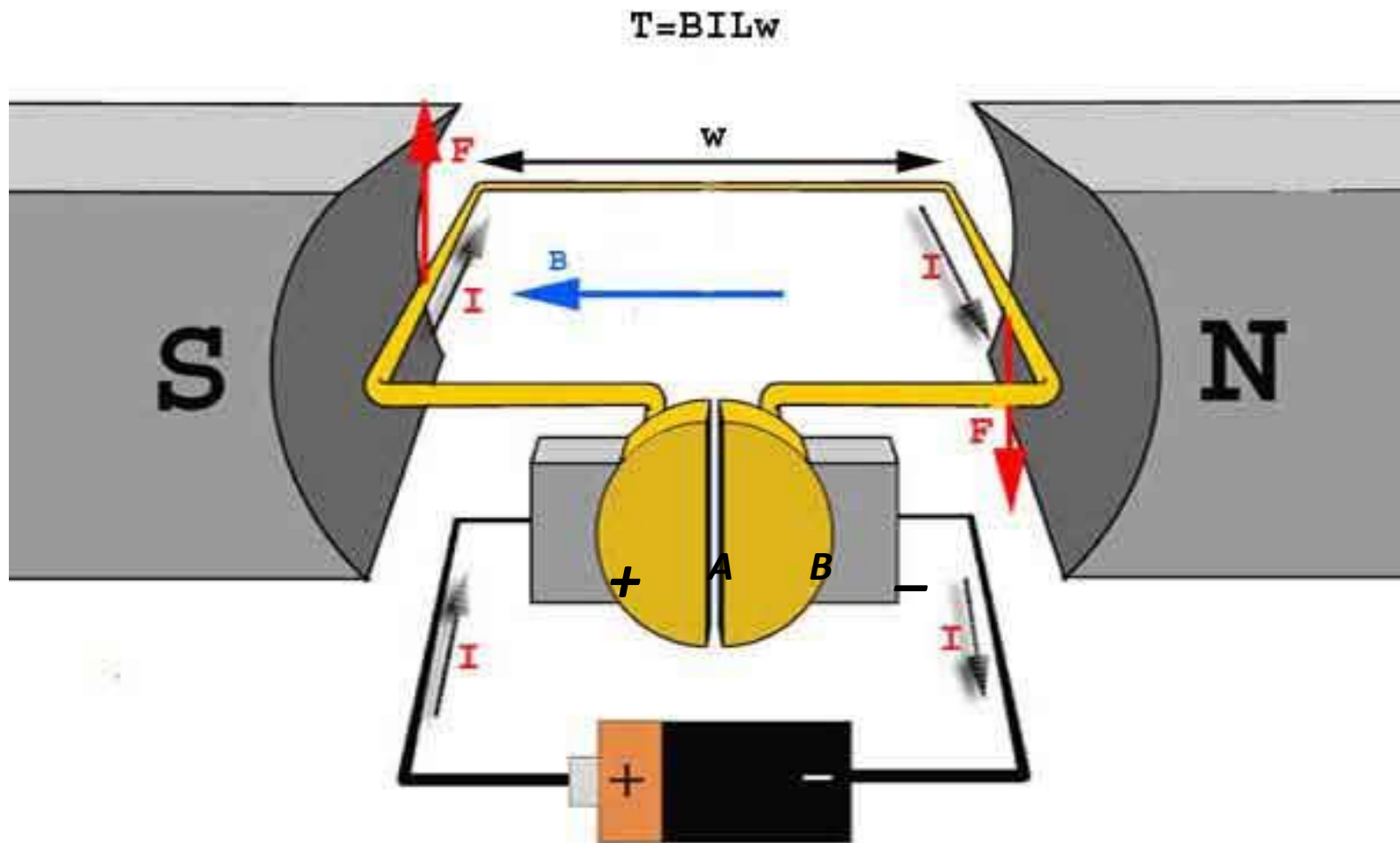
$$V_T = e_A$$

$$i_A = 0$$

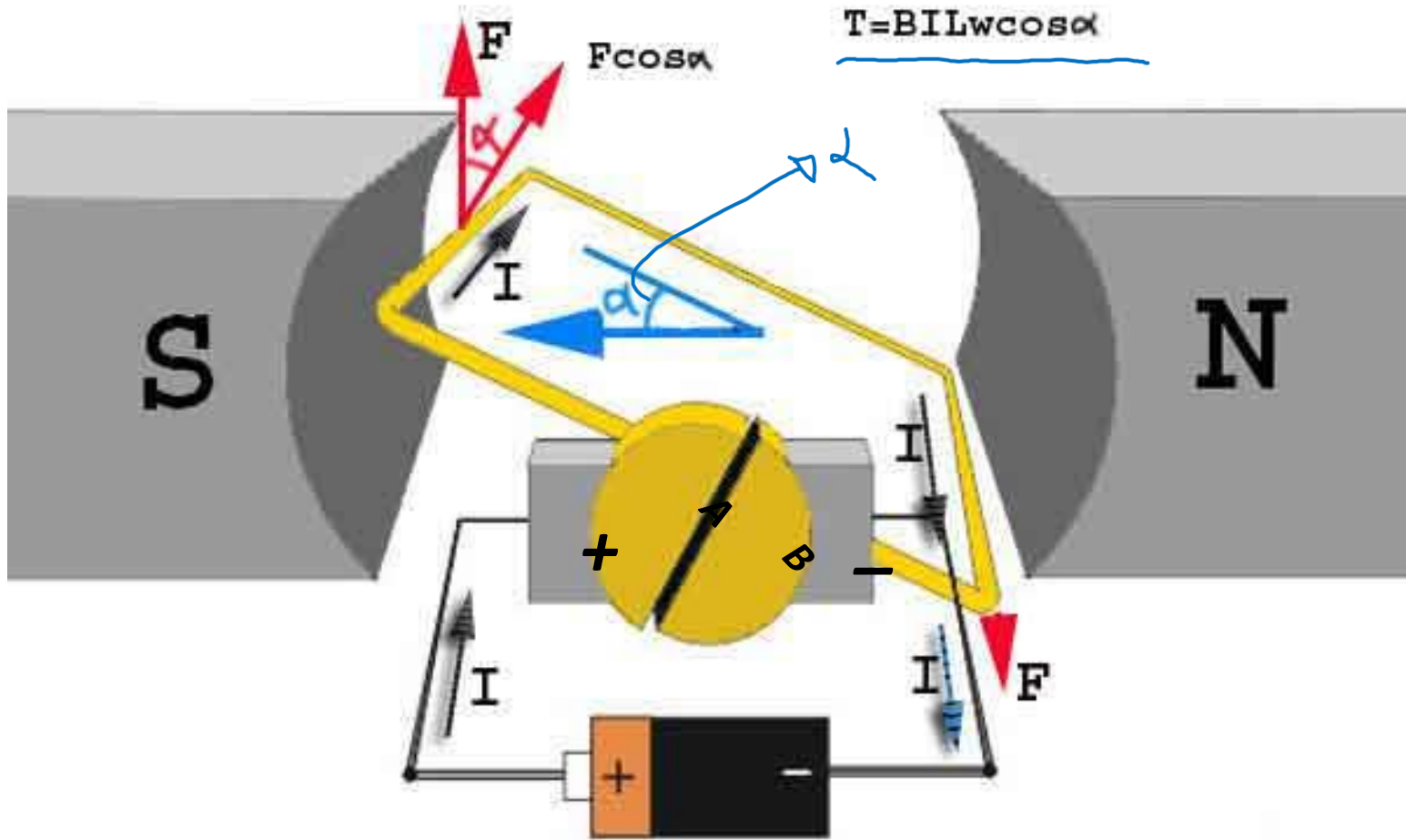
$$T = BILw \cos 90 = 0$$



Rotating DC Motor: Torque

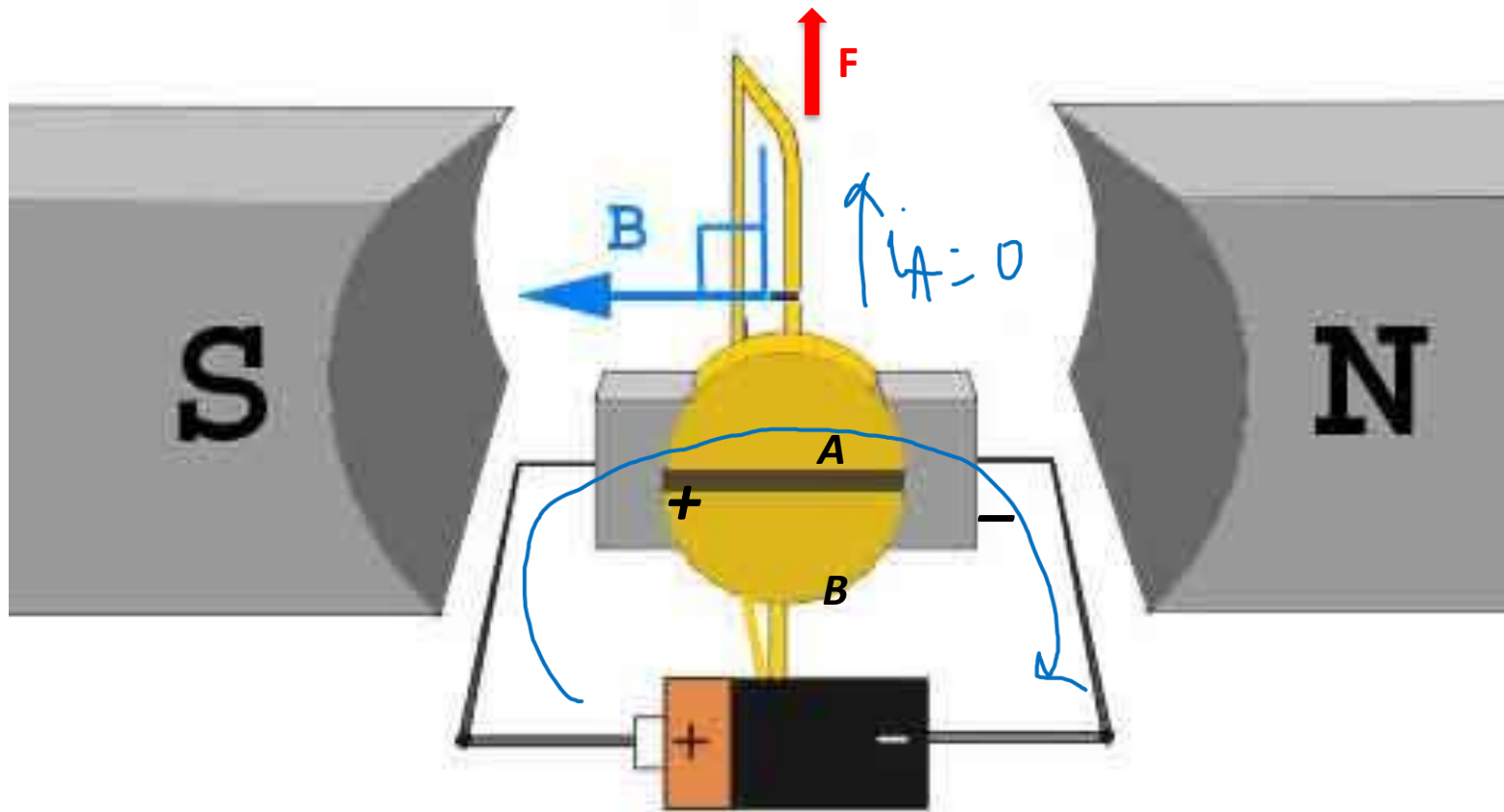


Rotating DC Motor

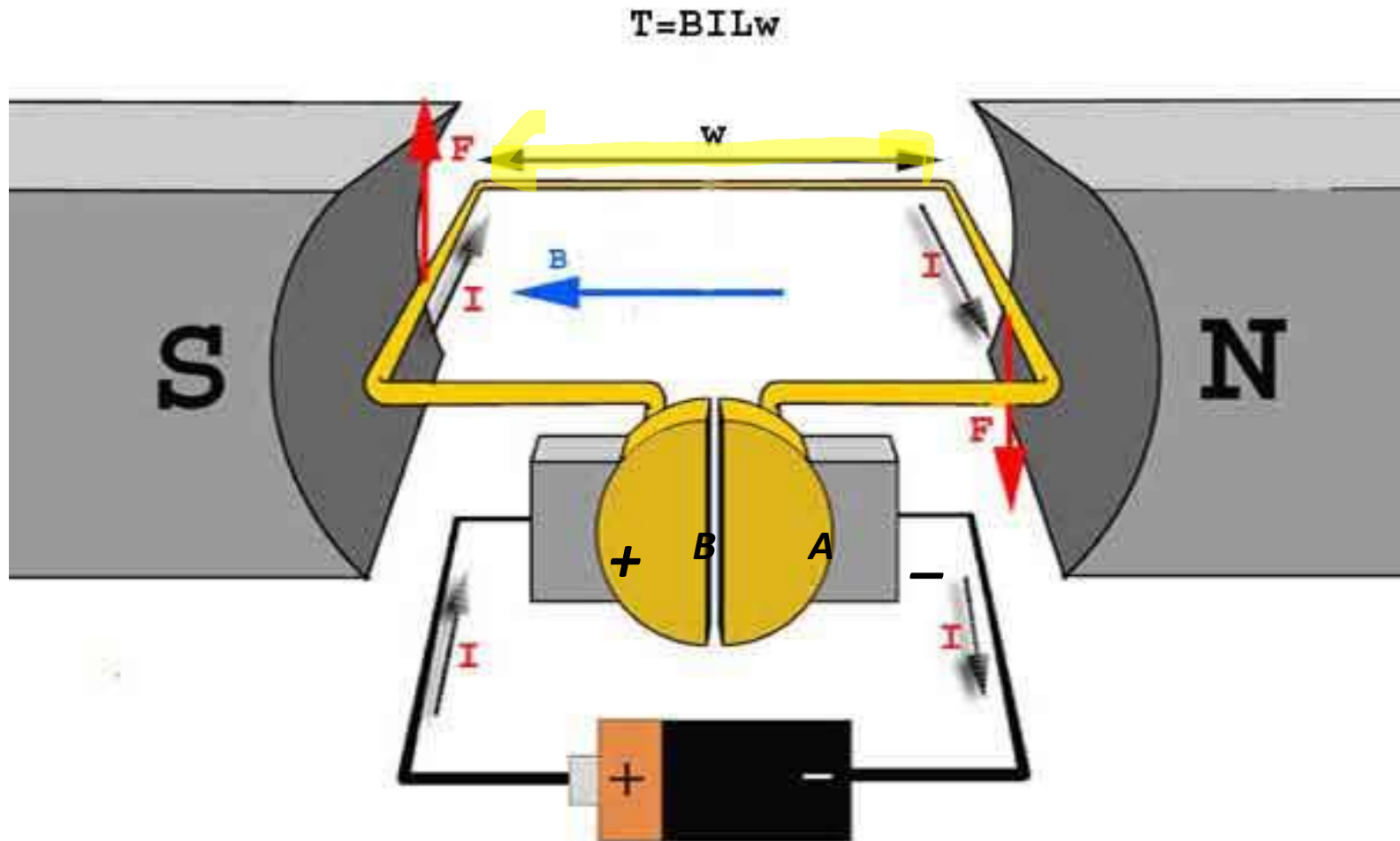


Rotating DC Motor

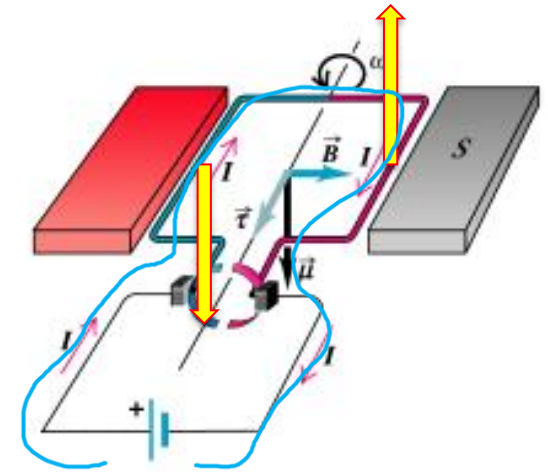
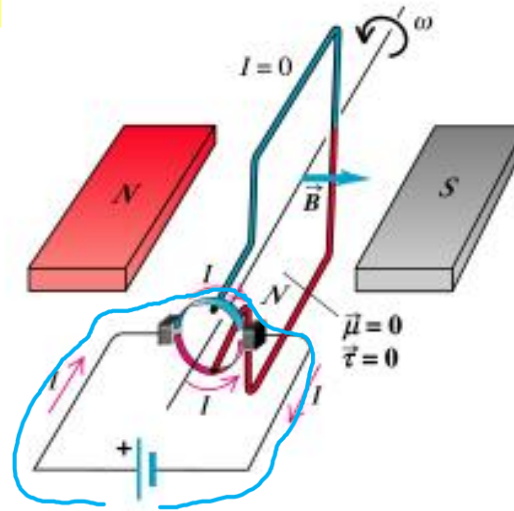
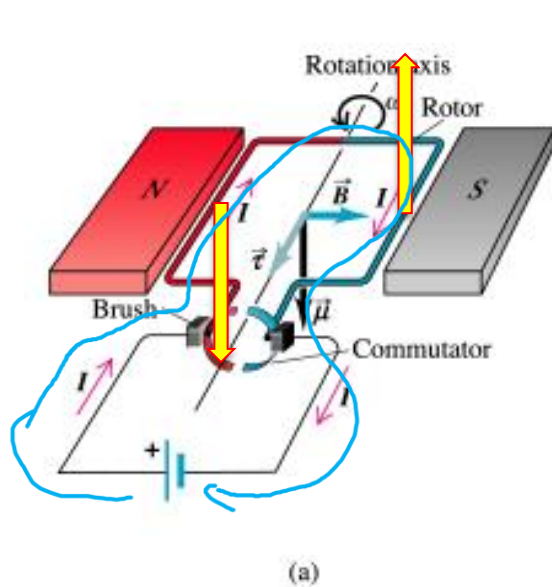
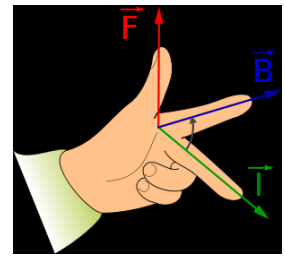
$$T = BILw \cos 90 = 0$$



Rotating DC Motor

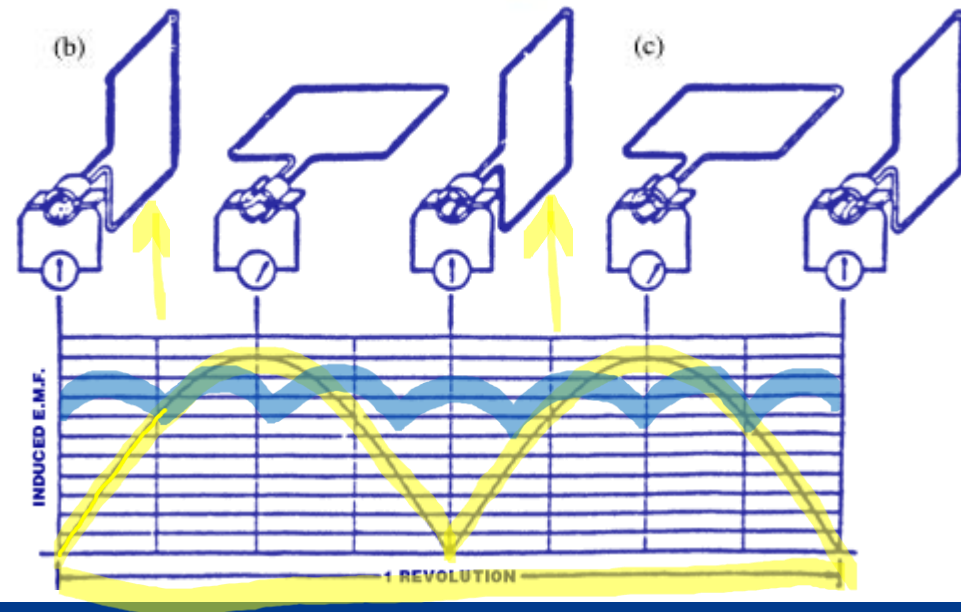


Back Electromotive Force (EMF)

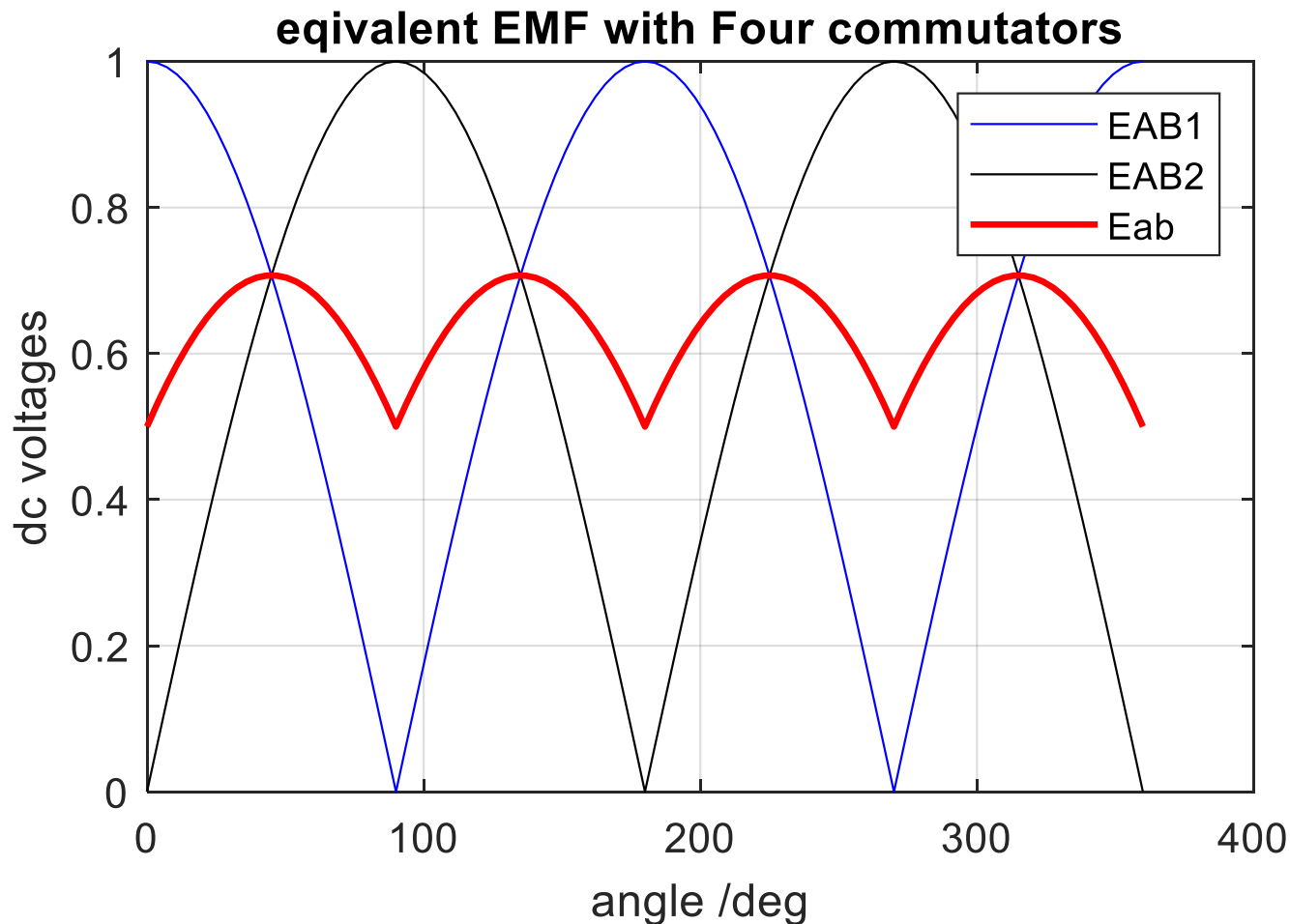


The stator and the rotor are made of iron to intensify the magnetic field

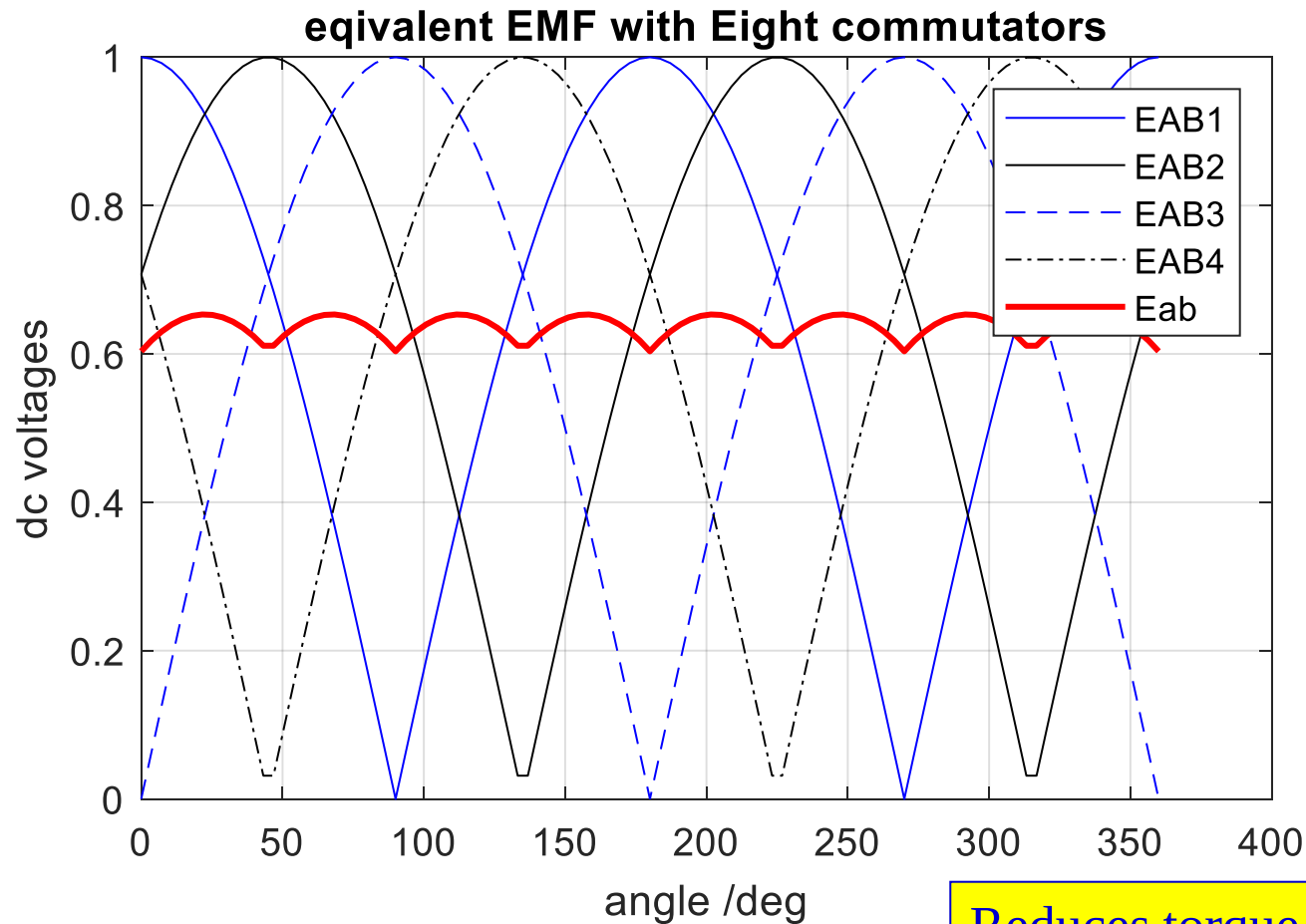
$$e = \frac{d\psi}{dt} = \frac{d(B \sin \alpha \cdot A)}{dt}$$



Reduced EMF Ripple with Multiple Commutators: Four (two pair)



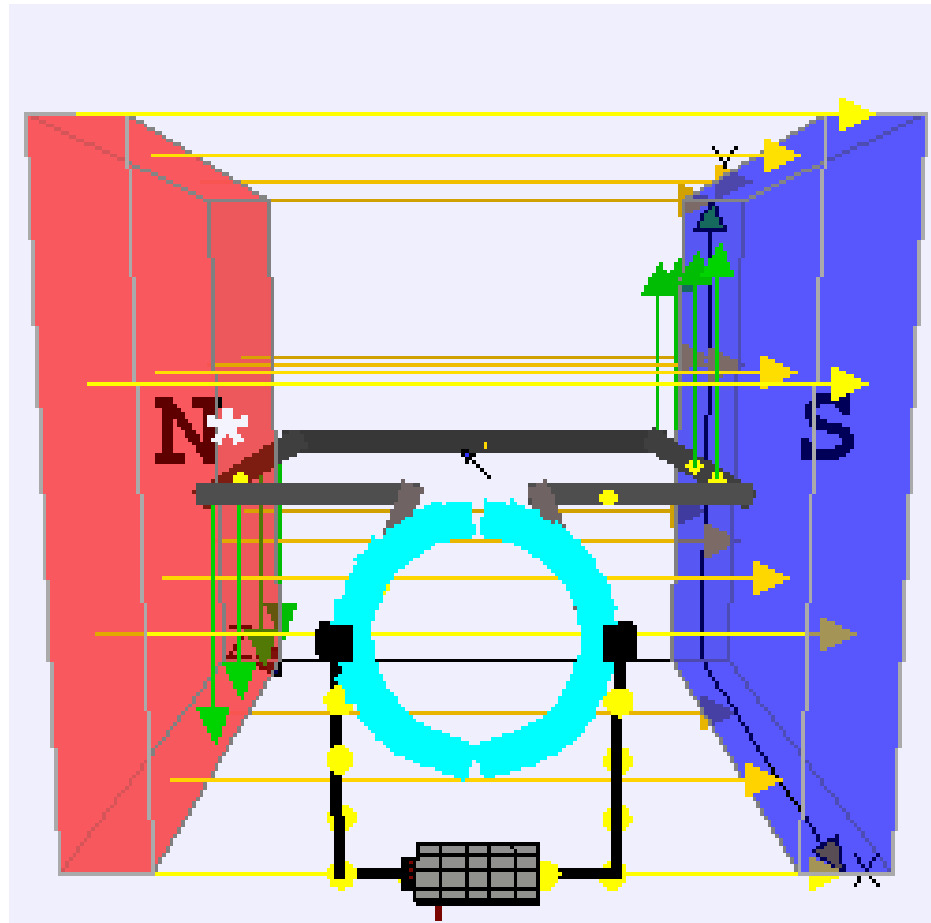
Reduced EMF Ripple with Multiple Commutators: Eight (four pair)



Reduces torque pulsation and vibrations

Animation

DC Motor working principle



Equivalent Circuit of DC Motors

Permanent-Magnet-based DC Motor

Field is set up by permanent magnets

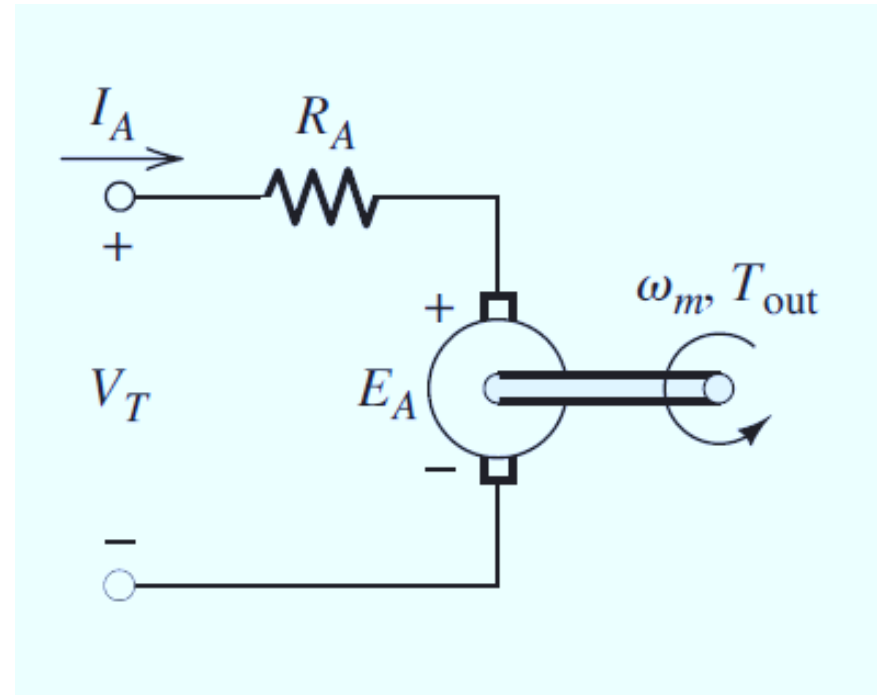
⇒ Approximately constant

Benefits:

- No Field losses
- Can be made very small

Dissadvantages:

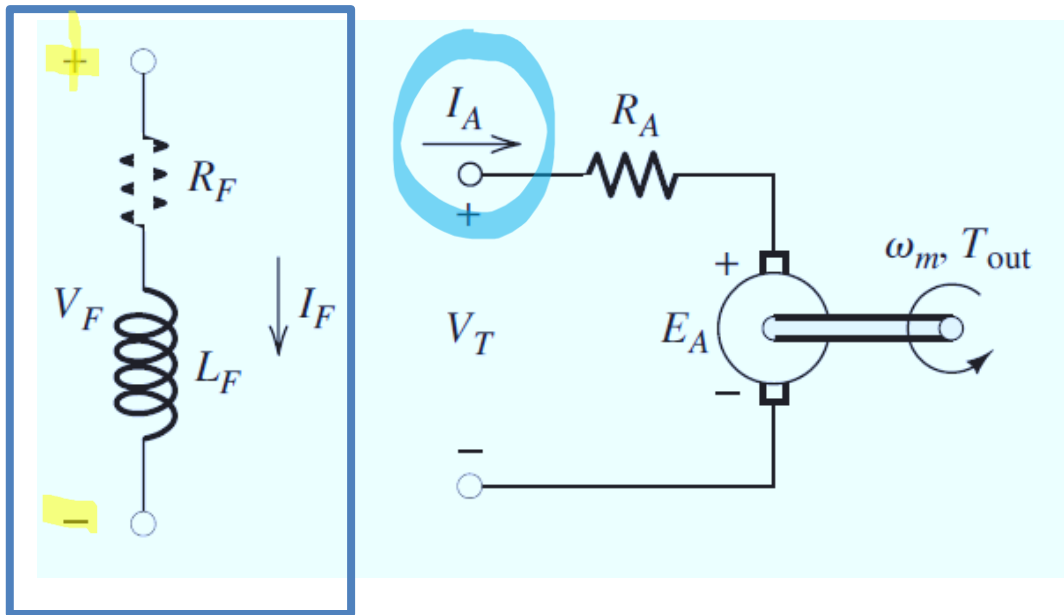
- Demagnetization over time (temperature)
- Weaker field
 - ⇒ Lower torque, higher speed than similar wound motors



$\phi(\text{flux}) = \text{constant}$
Provided by the permanent magnet

DC Motor Equivalent Circuit with External Excitation of Stator

Field circuit



$$e_A = E_A = \frac{d\Phi}{dt}$$

$$V_F = R_F I_F$$

$$E_A = K\phi\omega_m$$

$$T_{\text{dev}} = K\phi I_A$$

$$P_{\text{dev}} = \omega_m T_{\text{dev}}$$

$$P_{\text{dev}} = E_A I_A$$

Summary

The induced voltage in a DC Motor is given by

$$E_A = K\phi\omega_m$$

The voltage E_A is the average voltage induced in the armature due to the motion of the conductors relative to the magnetic field.

The developed Torque:

$$T_{\text{dev}} = K\phi I_A$$

The developed Power (converted to mechanical power):

$$P_{\text{dev}} = \omega_m T_{\text{dev}}$$

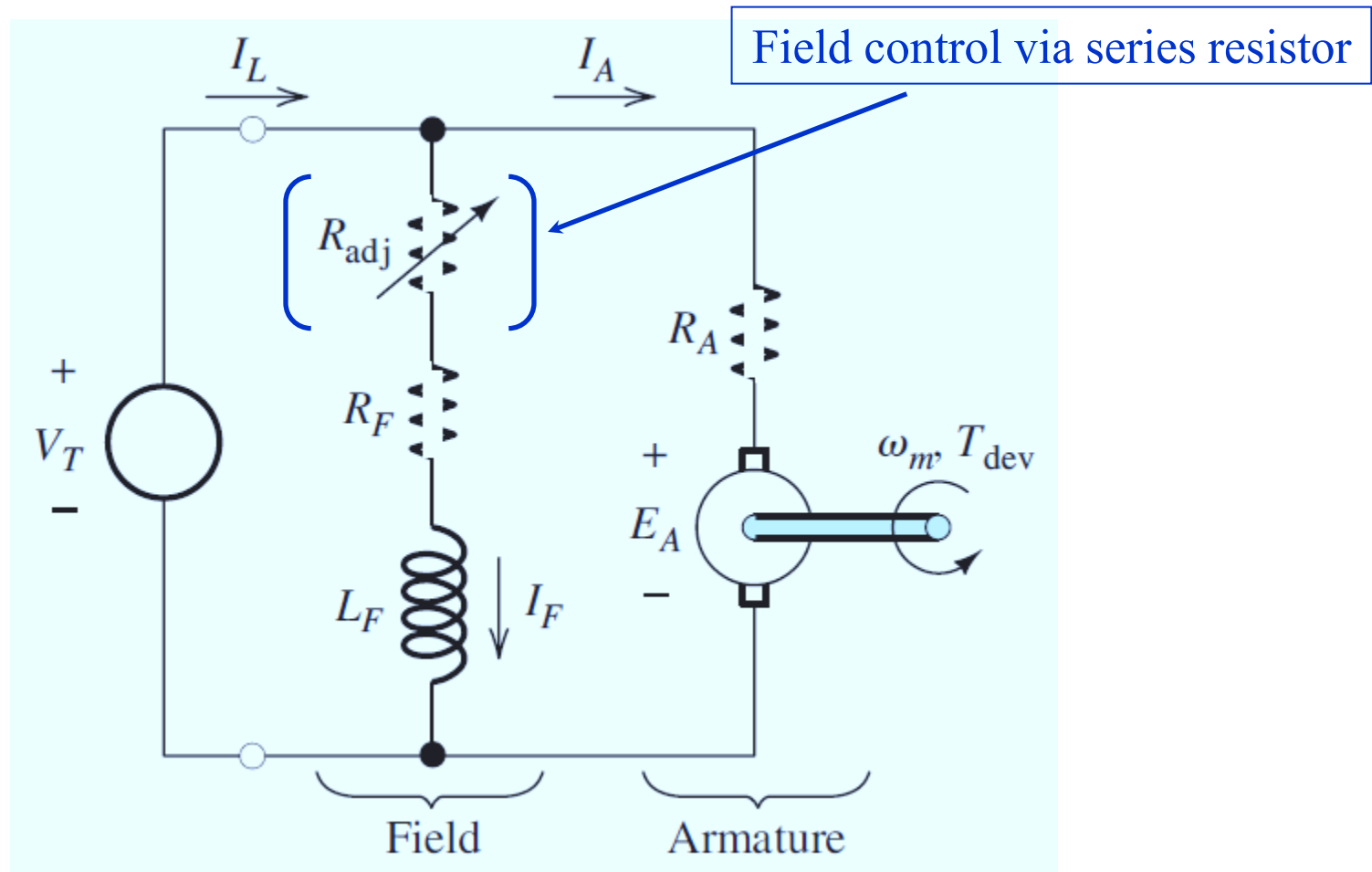
$$P_{\text{dev}} = E_A I_A$$

Energy Balance

Types of DC Motors

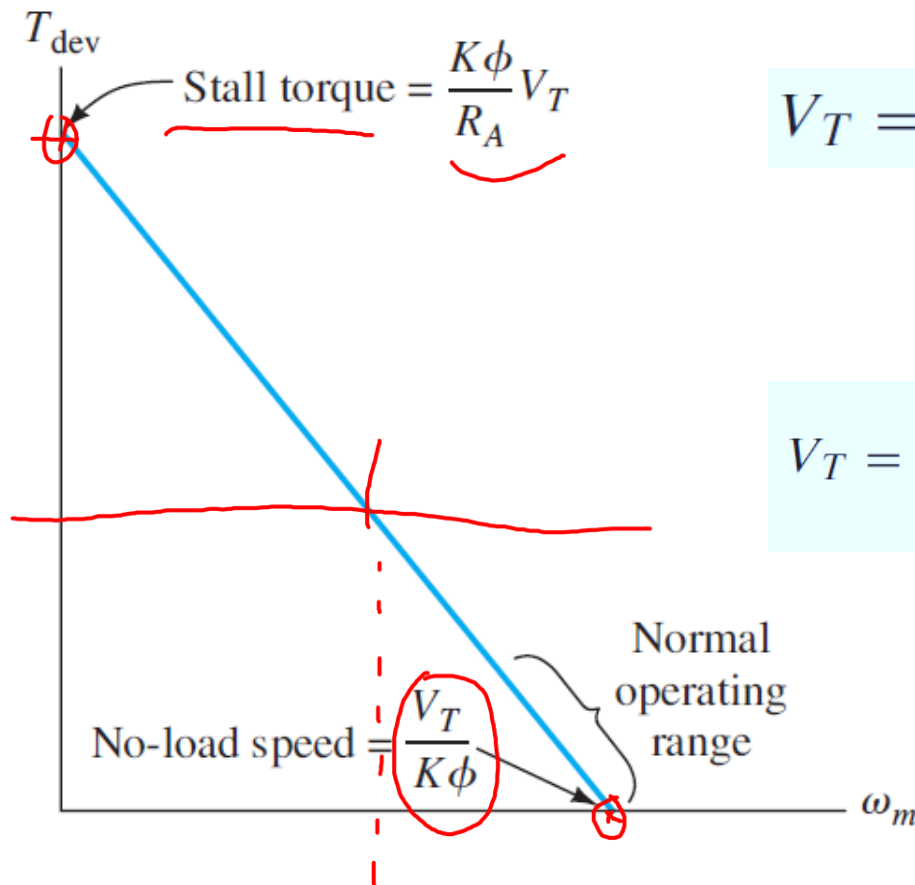
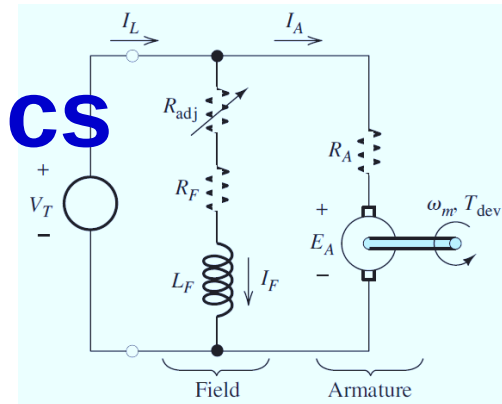
Parallelmotor («shunt motor»)

Field and armature winding connected in parallel



Torque-Speed Characteristics

Shunt DC Motor



$$V_T = R_A I_A + E_A$$

$$E_A = K\phi\omega_m$$

$$I_A = \frac{T_{\text{dev}}}{K\phi}$$

$$\phi = K_F I_F$$

$$V_T = \frac{R_A T_{\text{dev}}}{K\phi} + K\phi\omega_m$$

$$T_{\text{dev}} = \frac{K\phi}{R_A} (V_T - K\phi\omega_m)$$

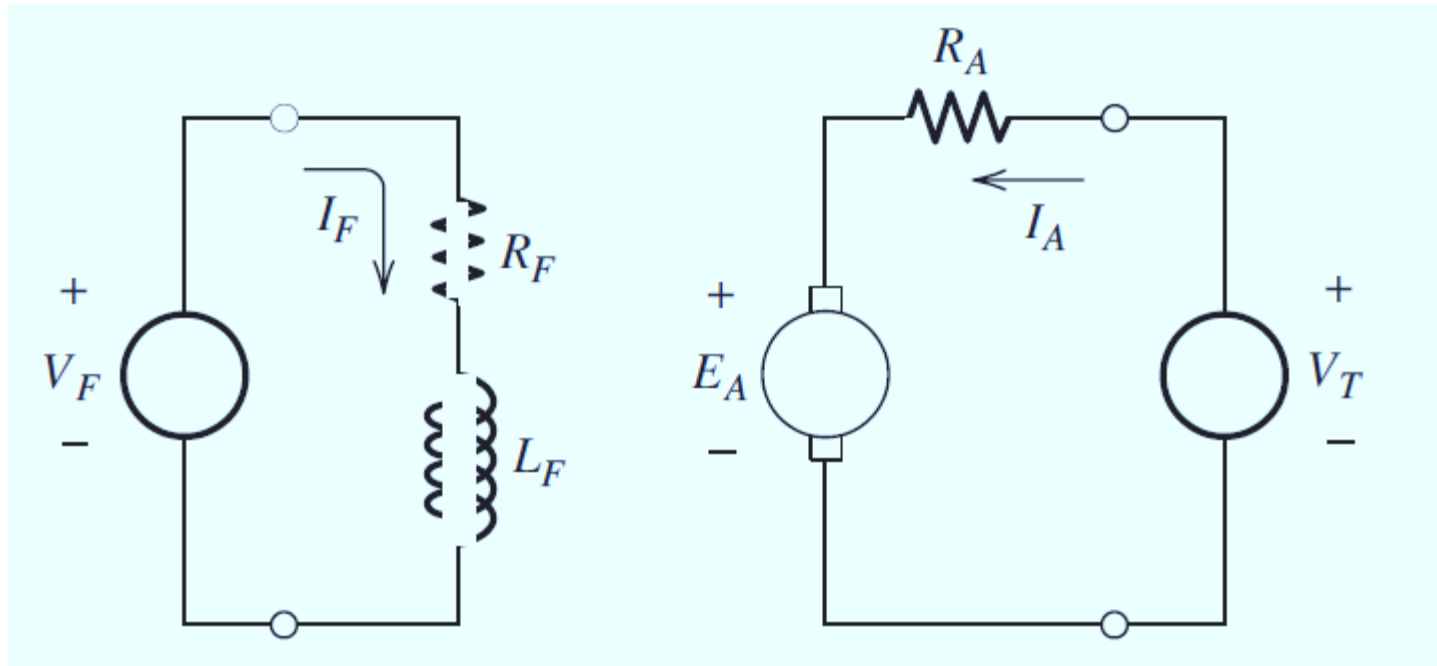
$$T_{\text{dev}} = 0 \quad V_T - K\phi\omega_m = 0$$

$$\omega_m = 0$$

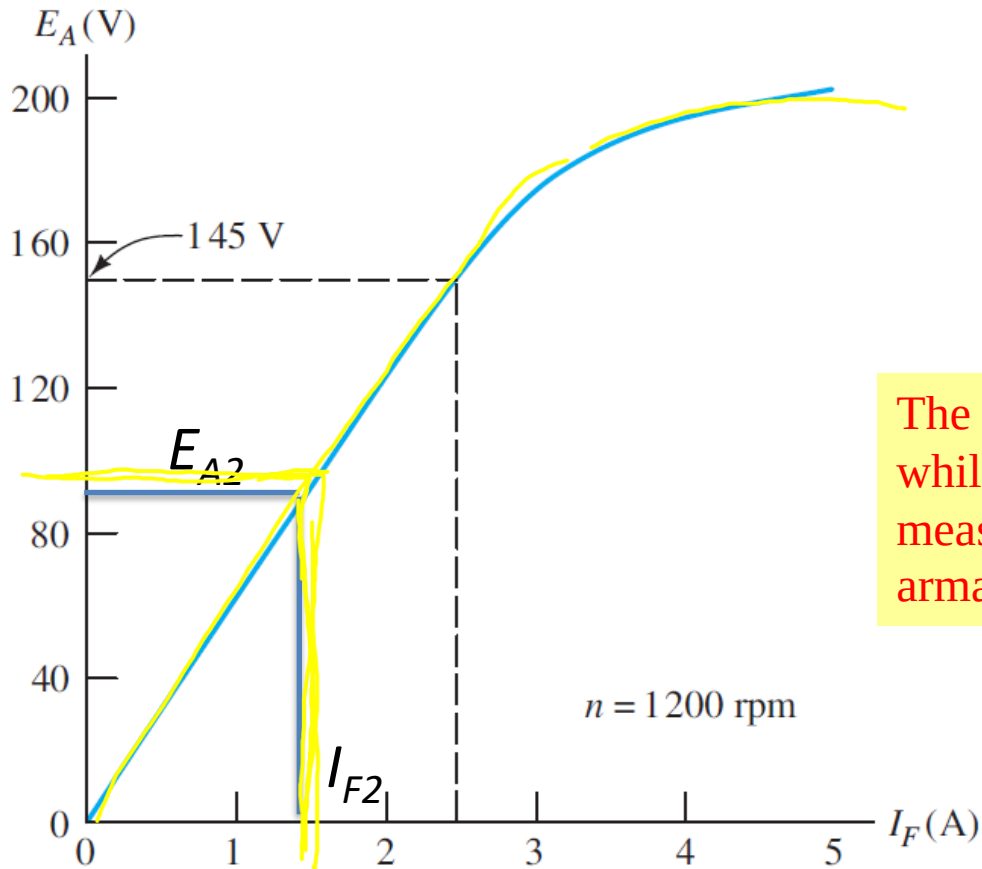
Separately Excited DC Motor

Two separate voltage sources, V_F , V_T

- Identical properties of shunt DC Motor
- Field voltage can be changed independent of armature voltage



Magnetization Curve



$$E_A = K\phi\omega_m$$

$$\phi = K_F I_F$$

$$\frac{E_{A1}}{E_{A2}} = \frac{K\phi\omega_1}{K\phi\omega_2}$$

The machine is driven at a constant speed while I_F is changed. E_A can be found by measuring the open-circuit voltage at the armature terminals.

$$\frac{E_{A1}}{E_{A2}} = \frac{n_1}{n_2} = \frac{\omega_1}{\omega_2}$$

The induced armature voltage E_A is directly proportional to speed. If E_{A1} represents the induced voltage at speed n_1 , and E_{A2} is the induced voltage at a second speed n_2

Example

The machine having the magnetization curve shown in Figure 16.15 is operating as a motor at a speed of 800 rpm with $I_A = 30$ A and $I_F = 2.5$ A. The armature resistance is $0.3\ \Omega$ and the field resistance is $R_F = 50\ \Omega$. Find the voltage V_F applied to the field circuit, the voltage V_T applied to the armature, the developed torque, and the developed power.

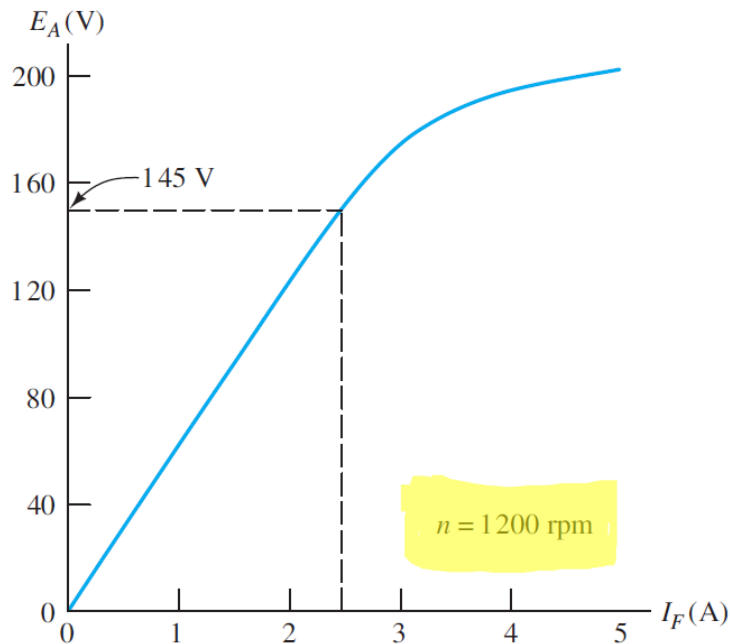
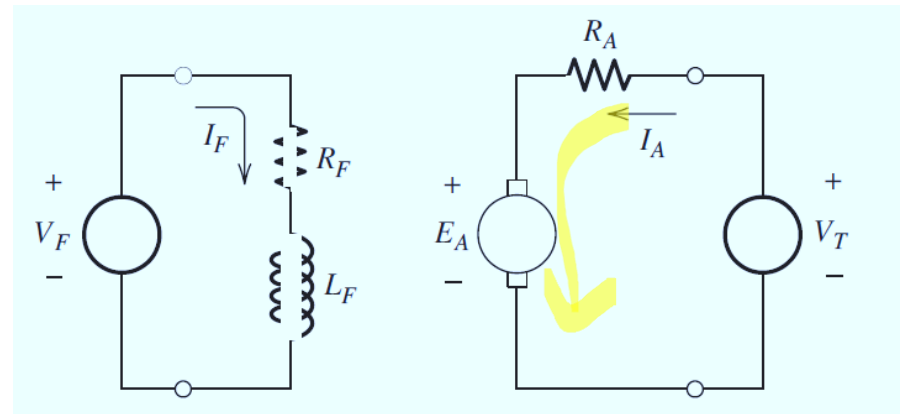


Fig.16.15



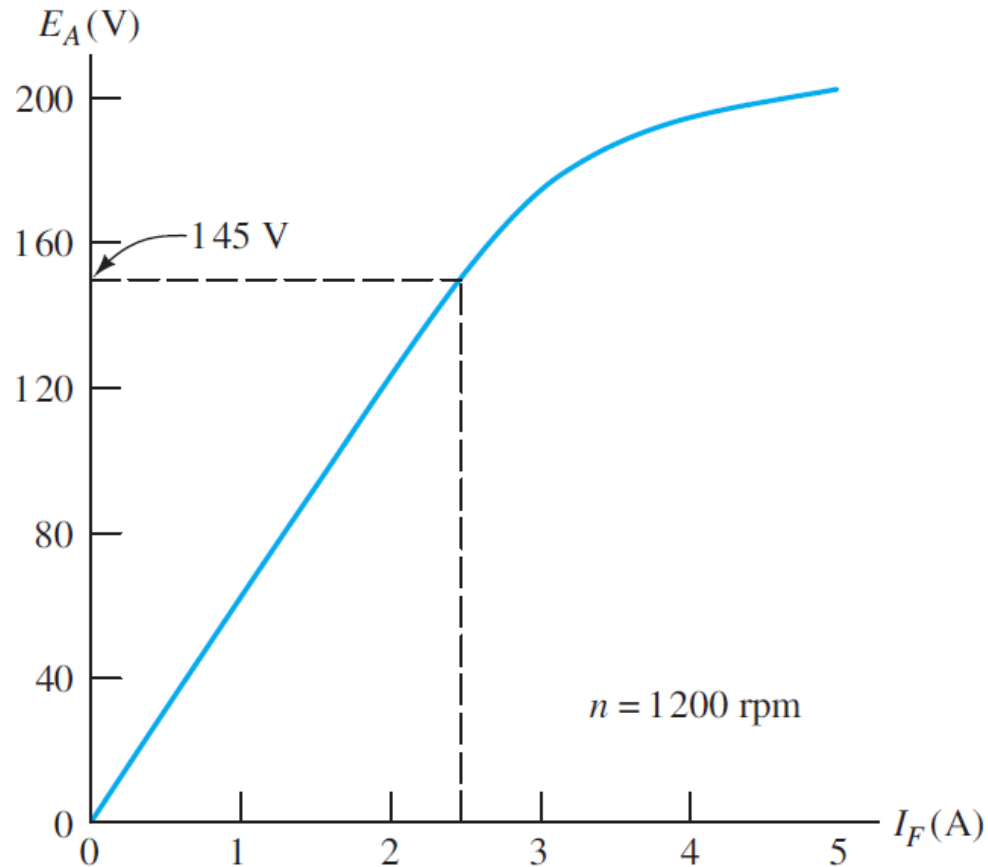
Equivalent circuit of the DC motor

Answers:

$$V_F = 125\text{V}, E_A = 96.67\text{V}, K\phi = 1.154$$

$$T_{dev} = 34.62\text{Nm}, P_{dev} = 2900\text{W}, V_T = 105.67$$

Magnetization Curve



$$E_A = K\phi\omega_m$$

$$\phi = K_F I_F$$

$$\frac{E_{A1}}{E_{A2}} = \frac{n_1}{n_2} = \frac{\omega_1}{\omega_2}$$

$$V_F = R_F I_F = 50 \times 2,5 = 125 \text{ V}$$

$$E_{A1} = 145 \text{ V} \text{ at } I_F = 2,5$$

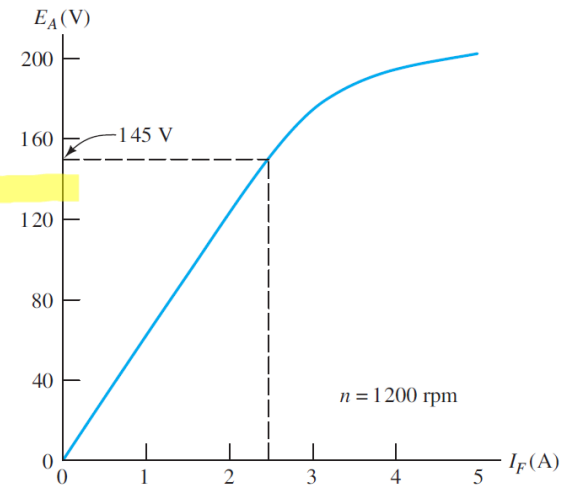
$$n_1 = 1200 \text{ rpm}$$

$$\text{At } n_2 = 800 \text{ rpm and } I_F = 2,5$$

$$E_{A2} = ?$$

$$E_{A2} = \frac{n_2}{n_1} \times E_{A1} = \frac{800}{1200} \times 145 = 96,67 \text{ V}$$

$$\omega_m = n_2 \times \frac{2\pi}{60} = 800 \times \frac{2\pi}{60} = 83,78 \frac{\text{rad}}{\text{s}}$$



$$E_A = K\phi \omega_m \rightarrow K\phi = \frac{E_A}{\omega_m}$$

$$K\phi = \frac{96,67}{83,78} = 1,154$$

$$I_A = 30 \text{ A.}$$

$$T_{dev} = K\phi I_A = 1,154 \times 30 = 34,62 \text{ Nm}$$

$$P_{dev} = \omega_m \cdot T_{dev} = 2900 \text{ W}$$

Verification of P_{dev} :

$$P_{dev} = I_A E_A = 30 \times 96,67 = \underline{\underline{2900 \text{ W}}}$$

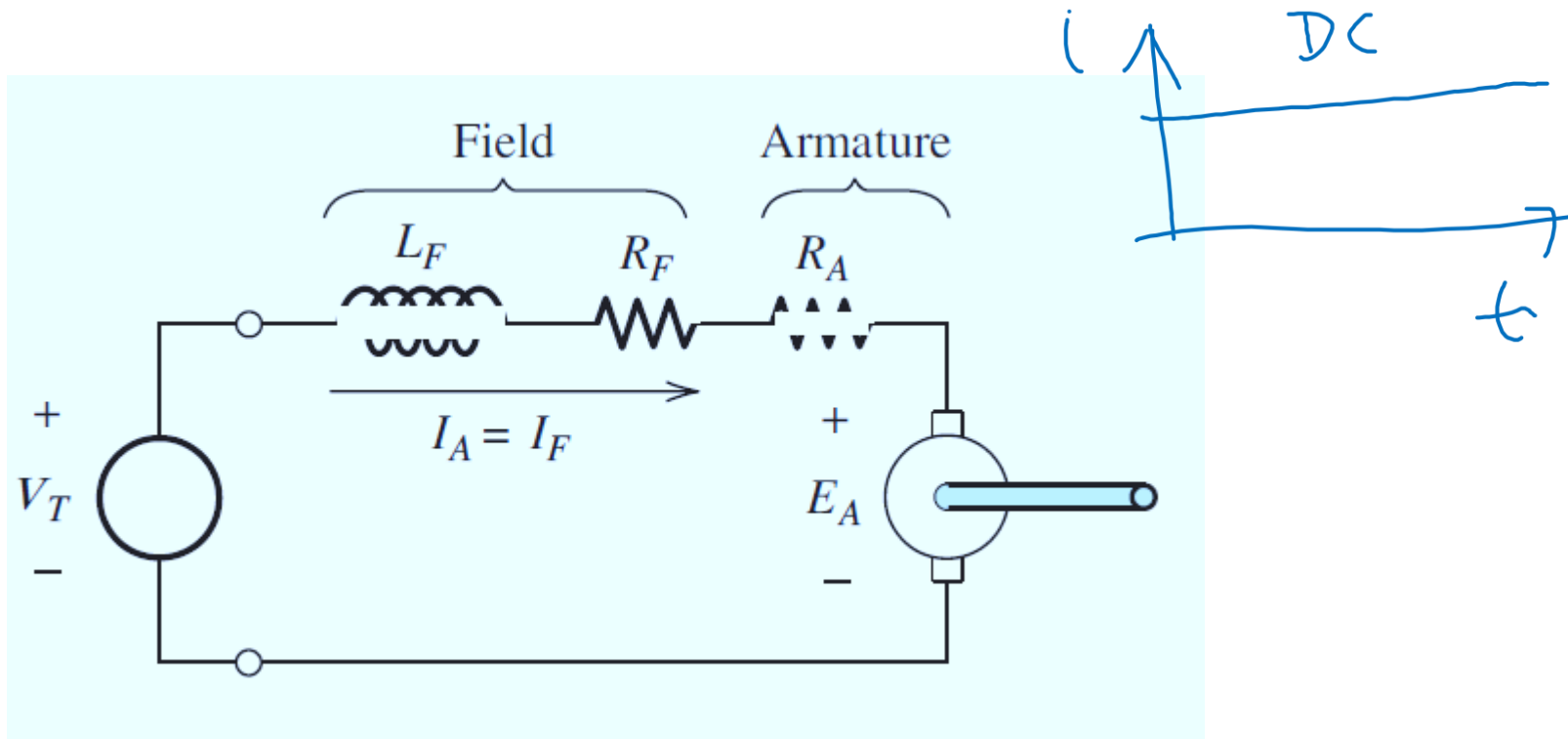
Battery voltage: V_T :

$$V_T = R_A I_A + E_A = 0,3 \times 30 + 96,67$$

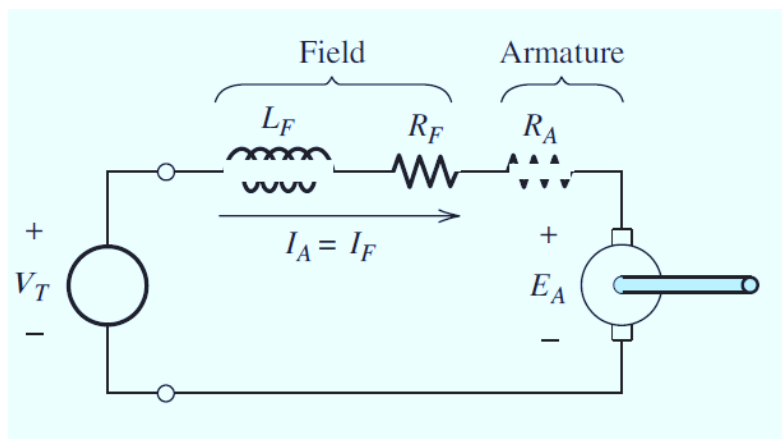
$$\underline{\underline{V_T = 105,67 \text{ V}}}$$

Series-connected DC Motors

Field and armature windings in series
Thick coils (low resistance) in field windings



Torque-Speed Characteristics



$$\phi = K_F I_F$$

$$E_A = K\phi\omega_m$$

$$\phi = K_F I_A$$

$$E_A = KK_F\omega_m I_A$$

$$T_{\text{dev}} = KK_F I_A^2$$

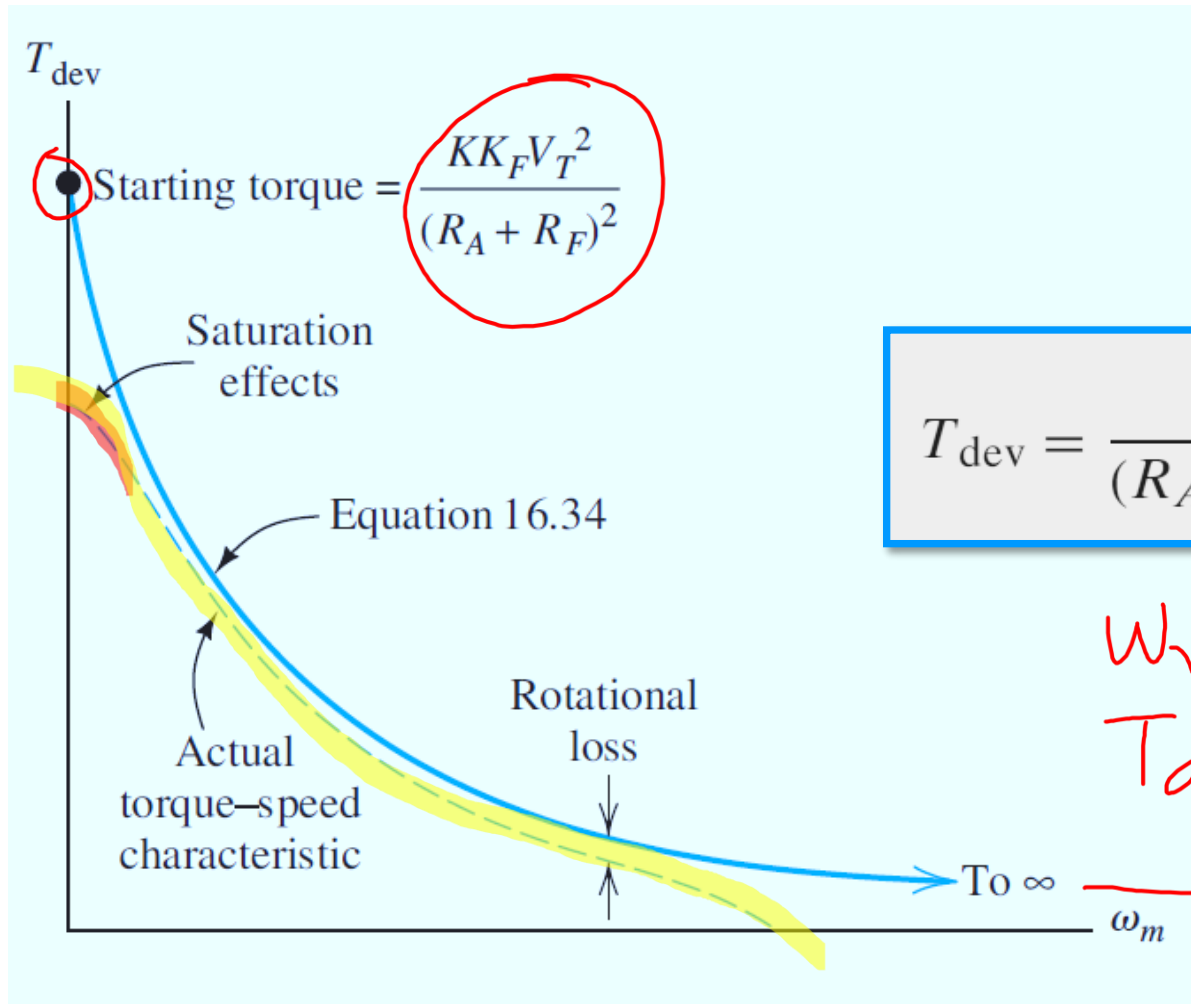
$$V_T = R_F I_A + R_A I_A + E_A$$

$$I_A = \frac{V_T}{R_A + R_F + KK_F\omega_m}$$

$$T_{\text{dev}} = \frac{KK_F V_T^2}{(R_A + R_F + KK_F\omega_m)^2}$$

Torque-speed
relation is non-linear

Torque-Speed Characteristics



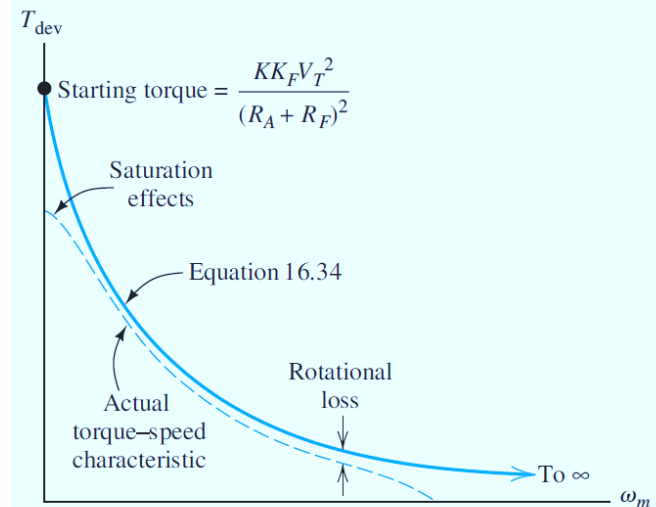
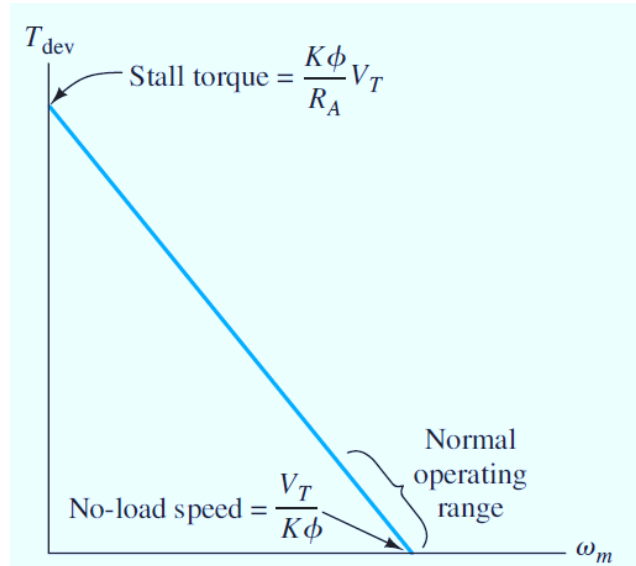
$$T_{dev} = \frac{KK_F V_T^2}{(R_A + R_F + KK_F \omega_m)^2}$$

$\omega_m = 0 \rightarrow T_{start}$
 $T_{dev} = 0 \rightarrow \omega_m = \infty$

Torque-Speed Characteristics

Shunt DC Motor
Externally Excited DC motor
Permanent-Magnet DC motor

Series DC Motor



End